

Notes on Algebraic Number Theory

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0. Introduction

This is a set of notes I am writing to help me study Algebraic Number theory. The main goal is just to speed through all the content as fast as possible so I can remember everything easily. I will also occasionally reference results from later in the textbook, since I have prioritized conceptual organization over a linear flow of results. Despite this I have taken care that there is no circular reasoning. I have also included some "problems", which are mainly just things I don't want to write out. A lot of the content is inspired by Professor Schläpke's course on algebraic number theory at UChicago. I must thank Professor Schläpke for giving such a good course.

"Wait, it's all lattices?"

1. Commutative Algebra

In this section I will speed through a lot of the basic commutative algebra and background knowledge needed to do Algebraic Number theory.

1.1. Fractional Ideals

Definition 1.1 (Fractional Ideals). Let R be an integral domain and K its field of fractions. A *fractional ideal* I is an R -submodule of K such that there exists some $r \in R$ with $rI \subseteq R$. We call I *invertible* if there exists a fractional ideal J with $IJ = R$.

In general it is hard to understand multiplication in interesting rings. However if we can instead understand multiplication in the *group of fractional ideals* we begin making progress.

Theorem 1.2.

- (a) If I is an invertible fractional ideal with $IJ = R$ then $J = R :_K I$.
- (b) We have the implications

$$I \text{ non-zero principal} \quad \Rightarrow \quad I \text{ invertible} \quad \Rightarrow \quad I \text{ fractional}$$

Proof.

- (a) By the hypothesis that I is invertible we see that $(R :_K I)$ must work as an inverse. Therefore it suffices to prove that inverses are unique. Suppose J, J' invert I and write:

$$J = JR = J(J'I) = J'(JI) = J'$$

- (b) A principal R -submodule of K , (x) has inverse (x^{-1}) . If I is an invertible fractional ideal then any element $x \in (R :_K I)$ must have $xI \subseteq R$ as desired.

□

In general for an R -submodule of K we will denote $(R :_K I)$ as I^{-1} .

Theorem 1.3. Any finitely generated R -submodule of K is a fractional ideal. Moreover if R is Noetherian then the opposite implication holds.

Proof.

$\Rightarrow$: Suppose that $I = (a_1/b_1, \dots, a_n/b_n)$ for $a_i, b_i \in R$. Then if $x = b_1 \cdots b_n$ we see $xI \subseteq R$.

$\Leftarrow$: By assumption we may find x with $xI \subseteq R$. Then by the Noetherian condition $xI = (a_1, \dots, a_n)$ so $I = (a_1/x, \dots, a_n/x)$ as desired. $\square$

Theorem 1.4. *Let I an invertible fractional ideal of R . Then I is finitely generated, and moreover if R is local then I is principal.*

Proof. Suppose that $II^{-1} = R$. Then since $1 \in II^{-1}$ we may find $a_i \in I$ and $b_i \in I^{-1}$ so that

$$\sum_{i=1}^n a_i b_i = 1$$

and $\langle a_1, \dots, a_n \rangle I^{-1} = R$. However by Theorem 1.2 I is the unique inverse to I^{-1} so $I = \langle a_1, \dots, a_n \rangle$ is finitely generated. In the case that R is local let $\mathfrak{m}$ be its maximal ideal and take modulo $\mathfrak{m}$ to see at least one of $a_k b_k$ is not in $\mathfrak{m}$. However since R is local this means that $a_k b_k = u$ is a unit. Therefore $\langle a_k \rangle I^{-1} = R$ so by the same argument $I = \langle a_k \rangle$. $\square$

Lemma 1.5. *If R is Noetherian and I is an ideal contained in the Jacobson radical then I^{-1} strictly contains R .*

Proof. Pick some $x \in I$. By Krull intersection theorem and the assumption let n be minimal so that $I^n \subseteq (x)$. Therefore $x^{-1}I^n \in R$, however $x^{-1}I^{n-1} \not\subseteq R$. Then any element $y \in x^{-1}I^{n-1} \setminus R$ satisfies $yI \subseteq R$ and $y \notin R$, showing that $R \subsetneq I^{-1}$. $\square$

1.2. Discrete Valuation Rings

Definition 1.6 (Valuation). A *valuation* on a ring R is a mapping $v : K \rightarrow G \cup \{\infty\}$, where G is a totally ordered abelian group, satisfying:

- (a) $v(a) = \infty \iff a = 0$
- (b) $v(ab) = v(a) + v(b)$
- (c) $v(a + b) \geq \min\{v(a), v(b)\}$
- (d) The image of v is nontrivial

We call v a *discrete valuation* if $G \cong \mathbb{Z}$.

It is worth noting some people do not enforce the last condition. We mainly do it to disallow the trivial valuation $v(R \setminus \{0\}) = 0$. To give some intuition for a valuation, here are some familiar examples:

- The p -adic valuation on $\mathbb{Q}$. For a prime p , $v_p(x)$ is the exponent of the highest power of p dividing x .
- The a -adic valuation on $k(x)$. For $a \in k$, $v_a(f)$ is the order of the zero (if positive) or pole (if negative) of f at $x = a$.
- The degree valuation on $k(x)$. Defined as $v_\infty(P/Q) = \deg(Q) - \deg(P)$.
- The Laurent series valuation. On $k((t))$, $v(f)$ is the smallest n such that the coefficient a_n of t^n is non-zero.
- The monomial valuation on $k(x, y)$. For a fixed weight $w = (w_1, w_2)$, $v(\sum a_{ij} x^i y^j) = \min\{i w_1 + j w_2 : a_{ij} \neq 0\}$.

In general, thinking of a valuation as a way to count the order of a zero or a pole of a function on a point gives very good intuition. The reason we care about valuations is their connection to the concept of a valuation ring.

Definition 1.7 (Valuation Rings). We call a ring R with fraction field K a *valuation ring* if one of the two equivalent conditions is satisfied:

- (a) For every $x \in K$ either x or x^{-1} is in R .
- (b) There exists a valuation on K such that $R = \{x \in K : v(x) \geq 0\}$

Proof.

(a) $\Rightarrow$ (b): We define the valuation group $G = R/R^\times$ to be the multiplicative group of R quotient the units. The valuation is $v(r) = \bar{r}$ and $v(0) = \infty$. We need to show three things:

G is totally ordered: We claim that the ideals of R are totally ordered by inclusion. This induces a total order on G by $\bar{a} > \bar{b}$ if $(a) \subset (b)$. Suppose FTSOC that I and J are ideals such that we can find some $x \in I \setminus J$ and $y \in J \setminus I$. By assumption either x/y or y/x is in R , so suppose WLOG that $x/y \in R$. Then $x = y \cdot (x/y)$ which implies $x \in (y) \subseteq I$, a contradiction.

$v(ab) = v(a) + v(b)$: This follows directly from the definitions.

$v(a + b) \geq \min(v(a), v(b))$: This translates to the statement that

$$(a + b) \subseteq \max((a), (b)) = (a, b)$$

where the equality uses the fact that either $(a) \subseteq (b)$ or $(b) \subseteq (a)$.

(b) $\Rightarrow$ (a): For $x \in K$ we either have $v(x) \geq 0$ or $v(x) \leq 0$. If $v(x) \geq 0$, then $x \in R$. If $v(x) \leq 0$, then $v(x^{-1}) = -v(x) \geq 0$, so $x^{-1} \in R$.

□

Definition 1.8 (Discrete Valuation Ring). We call a valuation ring *discrete* if its valuation is discrete.

The concept of a discrete valuation ring (often abbreviated as DVR) is extremely important in algebraic number theory. Therefore try very hard to internalize the concept via the following examples:

- The ring of p -adic integers, denoted $\mathbb{Z}_p$. The uniformizer is the prime number p , and the unique maximal ideal is $p\mathbb{Z}_p$.
- Formal power series rings, denoted $k[[t]]$ where k is a field. The uniformizer is the variable t . Elements are of the form $\sum_{n=0}^{\infty} a_n t^n$.
- The localization of $\mathbb{Z}$ at a prime p , denoted $\mathbb{Z}_{(p)}$. This consists of rational numbers a/b where $p \nmid b$. The uniformizer is p .
- The local ring of a curve at a smooth point. If C is a non-singular algebraic curve and $P \in C$ is a point, the local ring $\mathcal{O}_{C,P}$ is a DVR. The uniformizer is any local parameter at P .

It is worth noting that a ring having a valuation does not make it a valuation ring. However if R has a valuation, then $\mathfrak{p} = \{r \in R : v(r) > 0\}$ is a prime ideal of the subring $R' = \{r \in R : v(r) \geq 0\}$. It turns out that the localization $R'_{\mathfrak{p}}$ is a valuation ring with $v(a/b) = v(a) - v(b)$.

Problem 1.9. Prove that a discrete valuation ring is a Noetherian local domain

Now we have some characterizations of a discrete valuation ring.

Theorem 1.10. For a Noetherian local ring R the following statements are equivalent:

- (a) R is a discrete valuation ring.
- (b) R is principal ideal domain, but not field.
- (c) R is a 1-dimensional unique factorization domain.
- (d) R is a 1-dimensional normal domain.

(e) R is a 1-dimensional regular ring.

Proof. Through the proofs we denote $\mathfrak{m}$ as the unique maximal ideal of R .

(a) $\Rightarrow$ (b): By the Noetherian condition every ideal can be written as $I = (a_1, \dots, a_n)$. Then $I = (a_i)$ where a_i is the generator with minimal valuation. Since the valuation is nontrivial there must be non-unit element.

(b) $\Rightarrow$ (c): Since a PID is a UFD we only have to show that R is 1-dimensional. Suppose FTSOC that there is a prime $\mathfrak{p} \subsetneq \mathfrak{m}$. We can write $p = (\rho)$, $\mathfrak{m} = (\pi)$ and find some element x with $x\pi = \rho$. However since ρ generates a prime ideal it is irreducible, a contradiction.

(c) $\Rightarrow$ (d): It is a general result that every UFD is normal.

(d) $\Rightarrow$ (e): We will show that $\mathfrak{m}$ is principle by showing it is invertible, so consider $\mathfrak{m}\mathfrak{m}^{-1}$ (where $\mathfrak{m}^{-1} := (R :_K \mathfrak{m})$). By definition this is an ideal in R containing $\mathfrak{m}$, therefore it is either $\mathfrak{m}$ or R .

$\mathfrak{m}\mathfrak{m}^{-1} = \mathfrak{m}$: In this case every element $x \in \mathfrak{m}^{-1}$ satisfies $x\mathfrak{m} \subseteq \mathfrak{m}$ which implies by Nakayama lemma that x is integral over R and therefore in R by assumption. However this is a contradiction with Lemma 1.5.

$\mathfrak{m}\mathfrak{m}^{-1} = R$: In this case $\mathfrak{m}$ is an invertible ideal. Now by Theorem 1.4 every invertible ideal is principal as desired.

(e) $\Rightarrow$ (a): By Nakayama $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$ implies that $\mathfrak{m}$ is principal with some generator π . We claim that every element $x \in R$ is of the form $u\pi^n$ for $u \in R^\times$, at which point it follows that R is a DVR. By Krull intersection theorem let n be maximal so that $x \in (\pi^n)$ so $x = y\pi^n$. By maximality of n we see $y \notin (\pi)$, and since π is the unique maximal ideal this means y is a unit.

□

These might seem hard to remember, but the last 3 conditions essentially say the same thing. That a DVR looks like the functions near a smooth point on a curve. The slogan is that for a *curve* the following are the same:

- There is *one* codimension-1 direction through a smooth point.
- Near a smooth point there is a single local coordinate (a *uniformizer*).
- Every function has a well-defined *order of vanishing* which is just an integer.

This is exactly the discrete valuation story. Geometrically, the last three conditions in Theorem 1.10 are three different ways of saying

“near P the curve looks like one smooth branch, i.e. like a line.”

In dimension 1 these points of view collapse; in higher dimension they genuinely diverge.

Here is a useful mental dictionary for a local ring $R = \mathcal{O}_{C,P}$ of an integral curve C at a point P :

- *Regular (smoothness / tangent space)*. At a smooth point you can choose one local parameter t cutting out P . Algebraically this means the maximal ideal is generated by one element:

$$\mathfrak{m}_P = (t),$$

and infinitesimally there is only one direction along the curve:

$$\dim_{\kappa(P)}(\mathfrak{m}_P/\mathfrak{m}_P^2) = 1.$$

A singular point forces at least two independent first-order directions (or “too many tangent vectors”), which is why $\mathfrak{m}_P$ stops being principal.

- *Normal (no hidden functions / normalization)*. The normalization $\nu : \tilde{C} \rightarrow C$ is the operation that “pulls apart” branches and “fills in” missing integral functions. Saying R is normal means that locally nothing changes under normalization: there are no missing integral elements. On a curve, the only way to fail normality is precisely to have a singularity like a cusp or a node: there is some rational function that is integral over R which should be considered regular on the curve, but is not actually in R .
- *UFD (divisors become a single integer)*. On a smooth curve, any non-zero function f has a well-defined integer $\text{ord}_P(f)$ measuring how many times it vanishes at P (negative if it has a pole). The key point is that for a smooth point this is *one integer*, not a vector of integers. This corresponds to the fact that every ideal is just a power of (t) :

$$I \subseteq R \iff I = (t^n) \text{ for a unique } n \geq 0.$$

So locally there is a single “prime divisor” $[P]$, and everything factors as a multiple of it. If the point has multiple branches, you would instead get a *tuple* of orders (one per branch), and there is no longer a single generator controlling all vanishing.

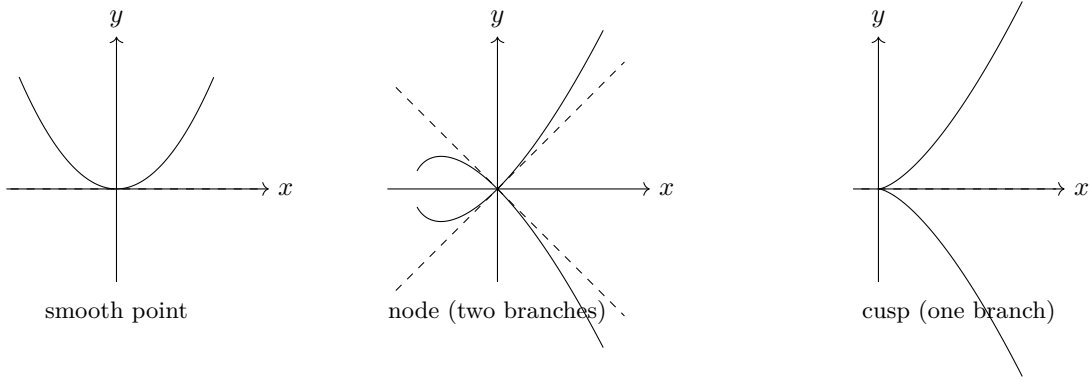


Figure 1: Singularities force “too many” first-order directions. At a smooth point there is one local parameter t ; at a node there are two branches; at a cusp there is one branch but the tangent is “thick”. In both singular pictures the maximal ideal cannot be generated by one element, and normalization changes the local ring.

It is helpful to keep two toy local models in mind (purely as intuition):

- *Cusp: one branch but not normal*. The parametrization $t \mapsto (t^2, t^3)$ suggests that (after completing) the local ring looks like

$$k[[t^2, t^3]] \subsetneq k[[t]].$$

The element t is “the missing uniformizer”: adjoining it is exactly what normalization does. This is why normality fails, and consequently regularity/UFD fail as well.

- *Node: two branches*. Near a node there are two ways to approach P along the curve, so vanishing wants to be measured by

$$(\text{ord}_{P,1}(f), \text{ord}_{P,2}(f)) \in \mathbb{Z}^2$$

rather than a single integer. Normalization separates the branches, and locally you should think of it as producing (roughly) two DVRs, one for each branch. The original local ring is their intersection inside the function field, which explains why it is not itself a DVR.

So, on a curve, *regular* says “one parameter” (one tangent direction), *normal* says “no hidden integral functions” (normalization does nothing), and *UFD* says “orders of vanishing are controlled by a single integer.” These are three shadows of the same geometric phenomenon.

1.3. Dedekind Domains

A DVR is a very local notion, so you might wonder if there is corresponding global notion. The answer is yes, and it is a Dedekind domain. Just like with a DVR, there are many equivalent conditions for a ring to be a Dedekind domain. Now buckle up, because we're about to start making the rounds.

Definition 1.11 (Dedekind Domain). We call an integral domain R that is not a field a Dedekind domain if it satisfies any of the equivalent conditions:

- (a) R is a Noetherian, and the localization at every maximal ideal is a discrete valuation ring.
- (b) R is a normal (or regular), 1-dimensional Noetherian domain.
- (c) Every nontrivial quotient R/I for $I \neq 0$ is a principal ideal ring.
- (d) Every nonzero proper ideal factors into primes.
- (e) Every nonzero fractional ideal of R is invertible.
- (f) For any ideals I, J in R , I is contained in J if and only if J divides I . That is, there is an ideal H with $I = JH$.

Proof.

(a) $\Leftrightarrow$ (b): Firstly, (a) implies that R is 1-dimensional since otherwise localizing at a maximal ideal above a nontrivial prime does not give a DVR. Recall from Theorem 1.10 that a DVR is a Noetherian, 1-dimensional, normal/regular *local* ring. Therefore we only need to show that normality and regularity are local properties, at least when every prime is maximal.

- Note that there is the standard inclusion map $\psi : R \rightarrow \bar{R}$ where $\bar{R}$ is the integral closure of R in its field of fractions. By commutative algebra ψ is an isomorphism iff $\psi_{\mathfrak{m}}$ is an isomorphism for all $\mathfrak{m} \in \text{Spm } R$. However since integral closure commutes with localization this is exactly the condition that $R_{\mathfrak{m}}$ is integrally closed for all $\mathfrak{m}$.
- Regularity means that $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$ for every $\mathfrak{m} \in \text{Spm } R$. However $R/\mathfrak{m}$ is local, and in particular $(R/\mathfrak{m})_{\mathfrak{m}} = R/\mathfrak{m}$ so this is a local condition.

(a) $\Rightarrow$ (d): Suppose that I is an ideal of R . Since R is Noetherian so by the Noether-Lasker theorem we may take a primary decomposition $I = Q_1 \cap \cdots \cap Q_n$. We want to show that $Q_i = \mathfrak{p}_i^k$ for the prime ideal $\mathfrak{p}_i = \sqrt{Q_i}$. We denote $S = R \setminus \mathfrak{p}$ and note:

$$(Q_i)_{\mathfrak{p}} \cap R = \{x \in R \mid (Q : x) \cap S \neq \emptyset\}$$

However by the primary condition if $sx \in Q_i$ for $s \notin \mathfrak{p}$ then $x \in Q_i$. Therefore $(Q_i)_{\mathfrak{p}} \cap R = Q_i$. Since $R_{\mathfrak{p}}$ is a DVR we see that $(Q_i)_{\mathfrak{p}} = \mathfrak{p}^k$ implying that $Q_i = \mathfrak{p}^k$. Since every prime is maximal by (b) we can use CRT to write this as $I = \mathfrak{p}_1^{k_1} \cdots \mathfrak{p}_n^{k_n}$

(d) $\Rightarrow$ (f): We claim that every prime ideal is maximal. The proof of this is long and in Jacobson's Basic Algebra. 1 However ignoring this technicality still constitutes a proof of (a) $\Rightarrow$ (f). Then for two ideals

$$I = \mathfrak{m}_1^{k_1} \cdots \mathfrak{m}_n^{k_n} \quad J = \mathfrak{m}_1^{r_1} \cdots \mathfrak{m}_n^{r_n}$$

we claim that $J \subseteq I$ iff $r_i \geq k_i$. This follows by localizing at $\mathfrak{m}_i$ to kill the other factors. Now $\mathfrak{m}_1^{k_1-r_1} \cdots \mathfrak{m}_n^{k_n-r_n}$ is J/I .

(f) $\Rightarrow$ (e): Firstly we claim integral ideals are invertible. Let I be a nonzero integral ideal of R , and choose any nonzero $a \in I$. Since $(a) \subseteq I$, (f) implies there exists an integral ideal H such that:

$$IH = (a)$$

Multiplying both sides by the fractional ideal (a^{-1}) yields $I(a^{-1}H) = R$. Since H is an integral ideal, $a^{-1}H$ is a fractional ideal. Thus, I is invertible.

Now let M be an arbitrary nonzero fractional ideal. By definition, there exists a nonzero $d \in R$ such that dM is an integral ideal. From Part 1, the integral ideal dM is invertible, meaning there exists a fractional ideal J such that:

$$(dM)J = R \implies M(dJ) = R$$

Since dJ is a fractional ideal, M is invertible.

(e) $\Rightarrow$ (a): Let $\mathfrak{m}$ be a maximal ideal of R and note that

$$I_{\mathfrak{m}} I_{\mathfrak{m}}^{-1} = (II^{-1})_{\mathfrak{m}} = R_{\mathfrak{m}}$$

so the localization retains the property of every ideal being invertible. Therefore $R_{\mathfrak{m}}$ is a PID by Theorem 1.4, so by Theorem 1.10 (a) we are done.

(a) $\Leftrightarrow$ (c): To prove (a) $\Rightarrow$ (c), note that any quotient decomposes according to the Chinese remainder theorem.

$$\frac{R}{\mathfrak{m}_1^{k_1} \cdots \mathfrak{m}_n^{k_n}} \cong \frac{R_{\mathfrak{m}_1}}{\mathfrak{m}_1^{k_1}} \cdots \frac{R_{\mathfrak{m}_n}}{\mathfrak{m}_n^{k_n}}$$

Each term is a quotient of a DVR, hence a PID by Theorem 1.10, and a product of PIDs is a PID.

In the other direction, we wish to show that $R_{\mathfrak{m}}$ is a PID for every $\mathfrak{m} \in \text{Spm } R$ by Theorem 1.10 (b). Consider two ideals $I, J \subseteq R_{\mathfrak{m}}$ and some $x \in IJ$. Then by the condition we may find some a, b with $I = (a, x)$ and $J = (b, x)$. However since $R/(x)$ is a local PIR we may assume $(a) \subseteq (b)$. Therefore $I \subseteq J$ as desired. Now write $I = (r, s)$ and note we may assume $(r) \subseteq (s)$. Therefore $I = (s)$ as desired.

□

In general the standard way to check whether a ring is a Dedekind domain is via (b). That is that a Dedekind domain is normal 1-dimensional Noetherian domain. One may think of (a)-(c) as expressing the notion that a Dedekind domain looks locally like a DVR. On the other hand (d)-(f) express the fact that the fractional ideals behave like a group. The connection is in algebraic geometry:

$$\text{fractional ideals} \quad \cong \quad \text{line bundles} \quad \cong \quad \text{divisors}$$

so being locally a DVR (smooth curve) means that divisors are just sums of points. As before, here are some examples of a Dedekind domain:

- The ring of integers $\mathbb{Z}$ is a principal ideal domain and the archetypal example of a Dedekind domain.
- The polynomial ring $k[x]$ over any field k is a Euclidean domain and therefore a Dedekind domain.
- The ring of integers $\mathcal{O}_K$ of a number field K (a finite extension of $\mathbb{Q}$) is a Dedekind domain.
- The Gaussian integers $\mathbb{Z}[i]$ form a principal ideal domain, which implies they are a Dedekind domain.
- Any discrete valuation ring (DVR), such as the ring of p -adic integers $\mathbb{Z}_p$, is a local Dedekind domain.
- The coordinate ring $k[C]$ of a smooth (non-singular) affine algebraic curve C over an algebraically closed field is a Dedekind domain.
- The ring $\mathbb{Z}[\sqrt{-5}]$ is a Dedekind domain that is not a principal ideal domain, illustrating that element factorization need not be unique even when ideal factorization is.

Now we can define the main exact sequence of study in algebraic number theory.

Construction 1.12. Let $\mathcal{O}_K$ be a Dedekind domain with fraction field K . We define the following objects:

$\mathcal{I}$: The *ideal group* $\mathcal{I}$ is the group of fractional ideals under multiplication.

Cl_K : The *class group* $\text{Cl}_K := \mathcal{I}_K / \mathcal{P}_K$ is the quotient of $\mathcal{I}_K$ by the *principal divisors* $\mathcal{P}_K$ which are the group of principal fractional ideals.

Then there is the following exact sequence

$$1 \rightarrow \mathcal{O}_K^\times \rightarrow K^\times \rightarrow \mathcal{I}_K \rightarrow \text{Cl}_K \rightarrow 1 \quad (1)$$

Proof. We will show exactness at each point.

- The map $\mathcal{O}_K^\times \rightarrow K^\times$ is by inclusion and the map $K^\times \rightarrow \mathcal{I}_K$ is by $x \mapsto (x)$. The elements which generate $\mathcal{O}_K$ (the zero element in $\mathcal{I}_K$) are exactly the units.
- The map $\mathcal{I} \rightarrow \text{Cl}_K$ is the quotient map, and the kernel are the principal ideals. These are exactly the ideals of the form (x) for $x \in K^\times$.

□

Remark 1.13. Note that

$$\mathcal{I}_K \cong \bigoplus_{\mathfrak{p} \in \text{Spec } \mathcal{O}_K} \mathbb{Z}$$

Proof. Any fractional ideal can be characterized by the power of each prime appearing in the decomposition. □

Remark 1.13 gives us a very good understanding of the ideal group. We will eventually better understand $\mathcal{O}_K^\times$ and Cl_K so that we may understand multiplication in $K^\times$ and $\mathcal{O}_K$ via the exact sequence in Eq. (1).

2. Ideals

In this section you will learn how to determine what primes do in extensions of Dedekind domains (such as number fields).

2.1. The Trace Pairing

In this section I will define the discriminant of a field extension, and how it comes from the trace pairing.

Definition 2.1 (Bilinear Form). A k -linear *bilinear form* on a vector space V is a bilinear map $V \times V \rightarrow k$. If V is finite dimensional, we define the *discriminant* of a bilinear form relative to a basis e_i as $\det \langle e_i, e_j \rangle$. We call a bilinear form *non-degenerate* or *perfect* if it satisfies the following equivalent conditions:

- (a) it has nonzero discriminant with respect to one (hence every) basis;
- (b) for every $x \in V$ there is a $y \in V$ with $\langle x, y \rangle \neq 0$;
- (c) for every $y \in V$ there is a $x \in V$ with $\langle x, y \rangle \neq 0$.

Proof. First we prove the claim that (a) does not care about basis. For a basis $e_1, \dots, e_n$ we let A denote the matrix $\langle e_i, e_j \rangle$. Then

$$\langle x, y \rangle = x^T A y$$

which can be verified by writing x and y in the e_i basis. Another basis $f_1, \dots, f_n$ induces an invertible change of basis matrix M with $M e_i = f_i$. Then

$$\langle f_i, f_j \rangle = \langle M e_i, M e_j \rangle = e_i^T M^T A M e_j$$

and the discriminant with respect to this basis is $\det M^T A M = \det A \cdot (\det M)^2$ which is nonzero iff $\det A$ is nonzero. Now by symmetry it suffices to prove (a) $\Leftrightarrow$ (c).

(a) $\Rightarrow$ (b): For some $x \in V$ we consider $y = A^{-1}x$ so that

$$\langle x, y \rangle = x^T A y = x^T x$$

which is always nonzero.

(b) $\Rightarrow$ (a): Suppose in the contrapositive that $\det A = 0$. Then there is some $y \in V$ with $Ay = 0$. It is now clear there is no $x \in V$ with $x^T Ay \neq 0$. □

Note that the discriminant of a bilinear form is defined up to the square of a number, due to the change of basis formula.

Remark 2.2. A good way to think about the discriminant is the following. A bilinear form $\langle \rangle$ gives some way to measure distance, angles, and volume. The discriminant of a basis $e_1, \dots, e_n$ is exactly the square of the covolume of the lattice it generates.

Proof. The discriminant with respect to an orthonormal basis $\langle e_i, e_j \rangle = \delta_{ij}$ is 1. Then changing the basis scales the discriminant by the square of the determinant of the change of basis matrix. □

Definition 2.3 (*R*-lattice). Let R be an integral domain with fraction field K , V a finite dimensional K -vector space, and M an R -submodule of V . We call a set of elements $\{\beta_1, \dots, \beta_n\}$ a *basis* for M if they generate M over R and are simultaneously a basis for V over K . M is called an *R-lattice* if it has a basis. We define the *discriminant* of M/R to be the discriminant of the trace pairing with respect to a basis of M .

Note that $\text{disc } V/K$ is an element of $K/K^{\times 2}$ (defined up to the square of an element) by our analysis in Definition 2.1. The field discriminant does not carry much information. On the other hand two bases of M over R must be related by an R -matrix, and $\text{disc } M/R$ is therefore an element of $R/R^{\times 2}$. This allows it to carry much more information since R might not have that many units. The most important example of this is when $R = \mathbb{Z}$ and M is the ring of integers in finite separable extension of $\mathbb{Q}$.

Lemma 2.4. Recall that the trace pairing defines an isomorphism $\tau_{V/R} : V \rightarrow V^\vee$ which identifies $M^\vee$ with $M^\sharp$ inside V . Then the discriminant is (up to units) the determinant of any lattice map sending $M^\sharp$ to M .

Proof. Let $e_1, \dots, e_n$ be a basis for M and $e_1^*, \dots, e_n^*$ be the dual basis generating $M^\sharp$. Since

$$e_i = \sum_j \langle e_i, e_j \rangle e_j^*$$

the matrix $\langle e_i, e_j \rangle$ is the change of basis matrix from $e_1, \dots, e_n$ to $e_1^*, \dots, e_n^*$. Therefore as a matrix in the dual basis it sends $M^\sharp$ to M . On the other hand the determinant is $\text{disc } M/R$ by definition. □

Construction 2.5. Let R be an integral domain with fraction field K , V a finite dimensional K -vector space, and $\langle \rangle$ a perfect pairing $V \times V \rightarrow K$. For any R -submodule M of V , we can define the *dual* of M as:

$$M^\sharp := \{x \in V : \langle x, y \rangle \in R \text{ for all } y \in M\}$$

Suppose moreover that M is an R lattice. Then $M^\sharp$ is an R -lattice generated by the *dual basis*.

Proof. Since the pairing is perfect it induces an isomorphism (via dimension counting)

$$V \xrightarrow{\sim} V^\vee \quad x \mapsto \langle x, - \rangle$$

and $M^\sharp$ is the preimage of $M^\vee = \text{Hom}_R(M, R)$ (sitting inside $V^\vee = \text{Hom}_K(V, K)$) under this isomorphism. Suppose that $e_1, \dots, e_n$ is an R -basis of M that is also a K -basis of V , we can use this isomorphism to induce a unique *dual basis* $e_1^*, \dots, e_n^*$ which has the property that:

$$\langle e_i, e_j^* \rangle = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

We claim that this dual basis $e_1^*, \dots, e_n^*$ generates $M^\sharp$. Suppose that $x \in M^\sharp$ and write $x = x_1 e_1^* + \dots + x_n e_n^*$ for $x_i \in K$. By the assumption that $x \in M^\sharp$ we see $\langle x, e_i \rangle = x_i$ is in R as desired. □

Geometrically suppose that $R = \mathbb{Z}$ in Construction 2.5. Then the conditions we put on M exactly translate to it being a full-rank *lattice* in $\mathbb{Q}^n$. The theorem tells us that the set of points whose pairing with the lattice gives only integers is another lattice. The *dual lattice* if you will. This geometric picture is very important to keep in mind.

Definition 2.6. Suppose that L/K is a finite extension, then L has the structure of a K vector space. Therefore any element $x \in L$ induces a K -linear transformation on L by multiplication by x . We can then define:

- the *norm* $N_{L/K}(x)$ to be the determinant of the transformation;
- the *trace* $\text{Tr}_{L/K}(x)$ to be the trace of the transformation.

Problem 2.7. Suppose $L/K/F$ is a tower of field extensions. Then:

$$N_{L/F} = N_{K/F} \circ N_{L/K} \quad \text{Tr}_{L/F} = \text{Tr}_{K/F} \circ \text{Tr}_{L/K}$$

Definition 2.8 (Trace Pairing). The *trace pairing* of a finite extension L/K is the bilinear form:

$$\langle x, y \rangle = \text{Tr}(xy)$$

The *discriminant* of the field extension is the discriminant of this pairing.

Proposition 2.9. Suppose that L/K is a degree n extension and that $\beta_1, \dots, \beta_n$ is a basis for L over K . Then

$$\text{Tr}(\beta_i \beta_j) = VV^T \quad V := \begin{bmatrix} 1 & \cdots & 1 \\ \sigma_1(\beta_1) & \cdots & \sigma_n(\beta_1) \\ \vdots & & \vdots \\ \sigma_1(\beta_n) & \cdots & \sigma_n(\beta_n) \end{bmatrix}$$

where V is the embedding matrix and the σ_i are the n embeddings of L into the algebraic closure of K . In particular, the discriminant of L/K is $(\det V)^2$.

Proof. We note that:

$$(VV^T)_{ij} = \sum_{k=1}^n V_{ik} V_{jk} = \sum_{k=1}^n \sigma_k(\beta_i) \sigma_k(\beta_j) = \text{Tr}(\beta_i \beta_j)$$

□

Corollary 2.10. The trace pairing of a finite extension L/K is non-degenerate if and only if the extension is separable.

Proof. By Proposition 2.9 it suffices to consider when $\det V$ is nonzero.

- If L/K is *separable*, the embeddings σ_i are distinct. Each one may be thought of as a homomorphism from the multiplicative group $L^\times$ to L . Therefore by Dedekind's theorem on the linear independence of characters $\det V \neq 0$.
- If L/K is *inseparable*, then $\sigma_i = \sigma_j$ for some $i \neq j$. This makes two columns of V the same so $\det V = 0$.

□

At this point the reader is encouraged to read Example 5.8 through Proposition 5.11. Now we will use the trace pairing to understand Dedekind domains.

Lemma 2.11. Let R be a normal domain with fraction field K , and let S be the integral closure of R in a separable extension L of K . There exist free R -submodules M and $M^\#$ with:

$$M \subseteq S \subseteq M^\#$$

Proof. Since L/K is separable, the trace pairing $(x, y) \mapsto \text{Tr}_{L/K}(xy)$ is non-degenerate. We may find a basis for L over K contained in S and let M be the R -module generated by this basis. Then Construction 2.5 tells us that

$$M^\# := \{x \in L : \text{Tr}(xy) \in R \text{ for all } y \in M\}$$

is finitely generated by the dual basis. Note that

$$\text{Tr}_{L/K}(SM) \subseteq \text{Tr}_{L/K}(S) \subseteq R$$

since $M \subseteq S$ and the trace of something integral over R is in R (by normality). Therefore $S \subseteq M^\#$ by definition and $M \subseteq S \subseteq M^\#$ as desired. $\square$

2.2. Prime Behaviour

You might ask why we just spent so much time talking about field theory, when understanding Dedekind domain is our main goal. That is due to the following theorem.

For the remainder let R be a Dedekind domain with fraction field K , let L/K be a finite separable extension of degree n , and let S be the integral closure of R in L .

Theorem 2.12. *S is a Dedekind domain.*

Proof. To prove that S is a Dedekind domain, we must verify three conditions:

- S is integrally closed in L . By definition, S is the integral closure of R in L . Since the integral closure of a ring in a field is always integrally closed, S is integrally closed in $L = \text{Frac}(S)$.
- S is Noetherian. Note that Lemma 2.11 implies that S is a submodule of a finitely generated module. Since R is Noetherian Hilbert's basis theorem implies that S is Noetherian.
- Every non-zero prime ideal of S is maximal. Let $\mathfrak{q}$ be a non-zero prime ideal of S , and let $\mathfrak{p} = \mathfrak{q} \cap R$. Since S is integral over R by definition, $\mathfrak{p}$ is a non-zero prime ideal of R . Because R is a Dedekind domain, $\mathfrak{p}$ is maximal. By the properties of integral extensions (specifically, the *Going-up Theorem*), the maximality of $\mathfrak{p}$ implies the maximality of $\mathfrak{q}$. Thus, S has Krull dimension 1. $\square$

Theorem 2.13. *If R is a Principal Ideal Domain then S is an R -lattice.*

Proof. Note that S is a finitely generated torsion-free R -module, therefore by the structure theorem for finitely generated modules over a PID $S \cong R^n$ as an R -module. We need to show the R -basis for S is also a K -basis for L . For linear independence, note that a linear dependence with coefficients in K leads to one with coefficients in R after clearing denominators. To show that this basis spans L , note any $x \in L$ satisfies a monic polynomial with coefficients in K . By clearing denominators we can find $r \in R$ with rx satisfying a monic polynomial with coefficients in R . Therefore $rx \in S$ as desired. $\square$

Definition 2.14. For any prime $\mathfrak{p} \in \text{Spec } R$ we may note

- (a) $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k}$;
- (b) The $\mathfrak{q}_i$ are exactly the primes with $\mathfrak{q}_i \cap R = \mathfrak{p}$.

We call the e_i the *ramification degree*, we call $\dim_{R/\mathfrak{p}} S/\mathfrak{q}_i$ the *inertia degree* of $\mathfrak{q}_i$ and denote it as $f(\mathfrak{q}_i|\mathfrak{p})$. We call the number of primes g . There are the following special cases:

- if $f(\mathfrak{q}_i|\mathfrak{p}) = e(\mathfrak{q}_i|\mathfrak{p}) = 1$ then $\mathfrak{p}$ is *totally split*;
- if $f(\mathfrak{q}_1|\mathfrak{p}) = g = 1$ then $\mathfrak{p}$ is *totally ramified*;
- if $e(\mathfrak{q}_1|\mathfrak{p}) = g = 1$ then $\mathfrak{p}$ is *inert*;

- if $e(\mathfrak{q}_i|\mathfrak{p}) = 1$ then $\mathfrak{p}$ is *unramified*, otherwise it is *ramified*.

Proof.

- (a) S is a Dedekind domain.
- (b) Note that $\mathfrak{q}_i$ contains $\mathfrak{p}S$ so by the prime avoidance theorem it must contain one of the factors in the decomposition. Then since every prime is maximal it is one of the factors.

□

Theorem 2.15. *For any nonzero prime ideal $\mathfrak{p} \subset R$, let the factorization of $\mathfrak{p}S$ in S be given by $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_g^{e_g}$, and let $f_i = [S/\mathfrak{q}_i : R/\mathfrak{p}]$ be the residue degree. Then:*

$$\sum_{i=1}^g e_i f_i = n$$

Proof. We analyze the dimension of the $R/\mathfrak{p}$ -vector space $S/\mathfrak{p}S$ in two different ways.

S is Locally Free: The localization $R_{\mathfrak{p}}$ is a DVR, and hence a PID. Then by Theorem 2.13 $S_{\mathfrak{p}}$ is an $R_{\mathfrak{p}}$ -lattice and $L \cong S \otimes_R K \cong S_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} K$, the rank of this free module is:

$$\dim_K(S_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} K) = [L : K] = n$$

Thus, S is locally free of rank n . This implies S is projective.

Dimension via Rank: We compute the dimension of $S/\mathfrak{p}S$ over the field $k = R/\mathfrak{p}$. Since S is locally free of rank n , we can localize at $\mathfrak{p}$ without changing the quotient:

$$S/\mathfrak{p}S \cong S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}} \cong S_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}S_{\mathfrak{p}}$$

Because $S_{\mathfrak{p}} \cong R_{\mathfrak{p}}^{\oplus n}$, we have:

$$S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}} \cong (R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}})^{\oplus n} \cong k^{\oplus n}$$

Therefore,

$$\dim_k(S/\mathfrak{p}S) = n$$

Dimension via Factorization: Alternatively, we use the prime factorization $\mathfrak{p}S = \prod_{i=1}^g \mathfrak{q}_i^{e_i}$. By the Chinese Remainder Theorem, since the powers $\mathfrak{q}_i^{e_i}$ are pairwise comaximal:

$$S/\mathfrak{p}S \cong \bigoplus_{i=1}^g S/\mathfrak{q}_i^{e_i}$$

The dimension is additive:

$$\dim_k(S/\mathfrak{p}S) = \sum_{i=1}^g \dim_k(S/\mathfrak{q}_i^{e_i})$$

Consider the filtration of $S/\mathfrak{q}_i^{e_i}$ given by the powers of $\mathfrak{q}_i$:

$$S \supset \mathfrak{q}_i \supset \mathfrak{q}_i^2 \supset \cdots \supset \mathfrak{q}_i^{e_i}$$

Each successive quotient $V_j = \mathfrak{q}_i^j / \mathfrak{q}_i^{j+1}$ is a vector space over the field $S/\mathfrak{q}_i$. Since $\mathfrak{q}_i$ is invertible in the Dedekind domain S , there are no proper ideals between $\mathfrak{q}_i^j$ and $\mathfrak{q}_i^{j+1}$, implying V_j is a one-dimensional vector space over $S/\mathfrak{q}_i$.

Thus, as a k -vector space:

$$\dim_k(\mathfrak{q}_i^j / \mathfrak{q}_i^{j+1}) = [S/\mathfrak{q}_i : k] \cdot \dim_{S/\mathfrak{q}_i}(V_j) = f_i \cdot 1 = f_i$$

Summing over the chain of length e_i , we get $\dim_k(S/\mathfrak{q}_i^{e_i}) = e_i f_i$.

Equating the two calculations of the dimension yields $n = \sum_{i=1}^g e_i f_i$. $\square$

Now that we understand what primes might do, it is natural to ask how we might figure out what a specific prime actually does in more detail. There are many tools for this which we will investigate in these sections. The most immediately natural one is the *Dedekind-Kummer* theorem.

Theorem 2.16. *Suppose that $L = K(\alpha)$ and let $f(x) \in R[x]$ be the minimal polynomial of α . If $\mathfrak{p}$ is a prime ideal of R such that there is a natural isomorphism:*

$$\phi : (R/\mathfrak{p})[x]/(\bar{f}(x)) \xrightarrow{\sim} S/\mathfrak{p}S \quad (1)$$

Then the factorization of $\bar{f}(x)$ into monic irreducibles in $(R/\mathfrak{p})[x]$ as $\bar{f}(x) = \bar{g}_1(x)^{e_1} \cdots \bar{g}_k(x)^{e_k}$ controls the prime factorization of $\mathfrak{p}S$ in the following way

$$\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k} \quad \mathfrak{q}_i := (\mathfrak{p}, g_i(\alpha)) \quad f(\mathfrak{q}_i|\mathfrak{p}) = \deg(\bar{g}_i)$$

where $g_i(x)$ is any lift of $\bar{g}_i(x)$ to $R[x]$.

Proof. We utilize the assumed structural isomorphism to transport the factorization from the polynomial ring to the Dedekind domain S .

Since the factors $\bar{g}_i(x)$ are distinct and irreducible in $(R/\mathfrak{p})[x]$, they are pairwise coprime. Applying the Chinese Remainder Theorem to the quotient ring $(R/\mathfrak{p})[x]/(\bar{f}(x))$ yields the decomposition:

$$\frac{(R/\mathfrak{p})[x]}{(\bar{f}(x))} \cong \prod_{i=1}^k \frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x)^{e_i})}$$

Via the isomorphism ϕ , we obtain the corresponding structure for the quotient:

$$S/\mathfrak{p}S \cong \prod_{i=1}^k \frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x)^{e_i})}$$

The prime ideals of S containing $\mathfrak{p}S$ correspond to the prime ideals of the quotient $S/\mathfrak{p}S$. Under the decomposition above, the maximal ideals of the product ring are generated by the classes of $\bar{g}_i(x)$.

Let $\mathfrak{q}_i$ be the ideal in S generated by $\mathfrak{p}$ and $g_i(\alpha)$. Under ϕ , the image of $\mathfrak{q}_i$ in $S/\mathfrak{p}S$ corresponds to the ideal $(\bar{g}_i(x))$. Since $(R/\mathfrak{p})[x]/(\bar{g}_i(x))$ is a field (as $\bar{g}_i$ is irreducible), the ideal $\mathfrak{q}_i$ is maximal in S . Thus, $\mathfrak{q}_1, \dots, \mathfrak{q}_k$ are the distinct prime ideals lying over $\mathfrak{p}$. The local factor $(R/\mathfrak{p})[x]/(\bar{g}_i(x)^{e_i})$ is a local ring with a unique maximal ideal which is nilpotent of index e_i . This implies that in the factorization of $\mathfrak{p}S$, the prime ideal $\mathfrak{q}_i$ must appear with exponent e_i . Therefore, $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k}$. Finally, the inertia degree is given by the dimension of the residue field extension:

$$f(\mathfrak{q}_i|\mathfrak{p}) = [S/\mathfrak{q}_i : R/\mathfrak{p}] = \dim_{R/\mathfrak{p}} \left(\frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x))} \right) = \deg(\bar{g}_i)$$

$\square$

While this theorem is very strong, and in particular tells us exactly how each prime behaves, the condition in Eq. (1) is very opaque. We will therefore spend some time talking about how to actually see whether this is true in the computational sections.

Problem 2.17. Understand quadratic extensions.

- Suppose that K is a field with characteristic not equal to 2. Prove that any degree 2 extension of K is generated by $K[\sqrt{d}]$ where $d \in R$ is squarefree.
- Suppose that $R = \mathbb{Z}$, $K = \mathbb{Q}$, and $L = \mathbb{Q}[\sqrt{d}]$ as in (a). Compute the ring of integers S , that is the integral closure of R in L .
- For every prime $p \in \mathbb{Z}$, determine whether it is inert, ramified, or split in the extension $L/\mathbb{Q}$.

2.3. Ideal Norms

In this section, I will define some machinery that will allow us to reason more smoothly with ideals. That is the ideal norm, a generalization of the norm.

Theorem 2.18. *An element $x \in S$ is a unit if and only if $N(x)$ is a unit.*

Proof.

$\Rightarrow$ In the case that x is a unit we have $N(x^{-1}) \in R$. Therefore

$$N(x)N(x^{-1}) = N(xx^{-1}) = N(1) = 1$$

implies that $N(x)$ is a unit.

$\Leftarrow$ By integrality of x write

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

to be the minimal polynomial of x and recall that $N(x)$ is a power of a_0 . Therefore $N(x)$ is a unit if and only if a_0 is a unit. If a_0 is a unit we can write

$$x^{-1} = \frac{-1}{a_0}(x^{n-1} + a_{n-1}x^{n-2} + \cdots + a_1)$$

which is an element of S .

□

Remark 2.19. If R is a PID recall Theorem 2.13 that S is an R -lattice. In this case multiplication by some $s \in S$ sends this lattice to a sub-lattice of itself via some R -matrix. The determinant of this matrix is by definition the norm of s (since the lattice is spanned by a K -basis of L).

By looking at the image of the lattice under s , we can detect the norm of s up to a unit. Essentially, we can see the ideal that the norm of s is generating. This motivates us to try to generalize the concept to ideals.

Definition 2.20. Let M be an R -lattice and N a sub-lattice. We call

$$[M : N]$$

the *index* of N in M and define it as the ideal generated by the determinant of any R -linear map sending M to N .

Proof. To prove this is well defined, choose maps A, B sending M to N . If N is not full-rank then $\det(A) = \det(B) = 0$. Otherwise choose a basis $M \cong R^n$ and consider the matrices $B^{-1}A$ and $A^{-1}B$. Since they both send M to M they have entries in R , hence determinants in R . However they are inverses of each other, implying that their determinants are units in R . Therefore $\det(A)/\det(B)$ is a unit, so $\det(A), \det(B)$ are the same up to a unit, and generate the same ideal as desired. □

The most important example of this is when $R = \mathbb{Z}$. In this case the determinant of the transition matrix is exactly the index of the sub-lattice

$$[M : N] = |M/N|$$

which motivates our definition.

Definition 2.21 (Ideal Norm). Suppose that R is a PID and that I is an ideal of S . Then I is a sub-lattice of S . We define the *ideal norm* of I as the index $[S : I]$.

Proof. There are a few things to check:

- This definition works since R being a PID does indeed make S an R -lattice;
- The ideal norm is never 0 since we may find some $x \in K$ such that $S \subseteq xI$. Therefore the sub-lattice must have full rank.

□

Unfortunately when R is not a PID this definition break down. Luckily the localization of a Dedekind domain is a PID (indeed, it is a DVR). Therefore we can compute the valuations of the ideal norm at each prime, and use that to characterize the full ideal norm.

Remark 2.22. The ideal norm agrees with the element norm on principal ideals

$$N((s)) = (N(s))$$

where $s \in S$.

Proof. The action of s on S (considered as an R -lattice) is an R -linear map sending S to (s) . □

Theorem 2.23. *The ideal norm is multiplicative. For ideals $\mathfrak{a}, \mathfrak{b} \subseteq S$:*

$$N_{L/K}(\mathfrak{a}\mathfrak{b}) = N_{L/K}(\mathfrak{a}) N_{L/K}(\mathfrak{b})$$

Proof. Working locally, we may assume that R is a DVR, therefore S is semi-local and a PID. Then for ideals (a) and (b) the result follows from the multiplicativity of the element norm

$$N(ab) = N(a) N(b)$$

and Remark 2.22. □

Lemma 2.24. *Let M be an R -lattice, N a full-rank sub-lattice, and D the index $[M : N]$. Then $DM \subseteq N$.*

Proof. Let A be a matrix sending M to N . Now by Cramer's rule:

$$\text{adj}(A)N = \text{adj}(A)AM = \det(A)M$$

If we choose a basis $M \cong R^n$ then A is an $n \times n$ R -matrix, since the adjugate of an R -matrix has entries in R it follows that $\text{adj}(A)N \subseteq N$. Therefore $\det(A)M \subseteq N$ and since this determinant is exactly the index (up to a unit) we are done. □

Corollary 2.25. *If I is an ideal of S then $I \cap R$ contains $N(I)$.*

Theorem 2.26. *The ideal norm is explicitly*

$$N(\mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_n^{e_n}) = \prod_i (R \cap \mathfrak{q}_i)^{e_i f_{\mathfrak{q}_i}}$$

where $f_{\mathfrak{q}_i}$ is the inertial degree of $\mathfrak{q}_i$.

Proof. By Theorem 2.23 it suffices to prove that $N(\mathfrak{q}) = \mathfrak{p}^f$ where f is the inertia of $\mathfrak{q}$ and $\mathfrak{p} = \mathfrak{q} \cap R$. Working locally, let R be a DVR, $\mathfrak{q} = (\lambda)$, and $\mathfrak{p} = (\pi)$. Write the exact sequence

$$S \xrightarrow{m_\lambda} \mathfrak{q} \rightarrow S/\mathfrak{q} \rightarrow 0$$

where m_λ is the matrix describing the multiplication by λ . By changing basis, we may put m_π into *Smith normal form* so that:

$$m_\lambda = \begin{bmatrix} \pi^{c_1} & 0 & \cdots & 0 \\ 0 & \pi^{c_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \\ 0 & 0 & 0 \cdots & 0 & \pi^{c_k} \end{bmatrix} \quad S/\mathfrak{q} \cong \prod_{i=1}^k R/\mathfrak{p}^{c_i}$$

From this expression we see that

$$\det m_\lambda = (\pi)^{c_1 + \cdots + c_k} = \mathfrak{p}^{c_1 + \cdots + c_k} \quad f_{\mathfrak{q}} = \dim_{R/\mathfrak{p}} S/\mathfrak{q} = c_1 + \cdots + c_k$$

as desired. □

It is worth noting that all this machinery also works for fractional ideals just as well. Now we will prove one last technicality.

2.4. The Different Ideal

In this section, we will investigate further prime ramification. We will need the assumption that $R/\mathfrak{q}$ is a perfect field (which is always true for number fields), in order to proceed.

Lemma 2.27 (Nilpotents force degeneracy). *Let k be a field and A a finite-dimensional commutative k -algebra. If A has a nontrivial nilpotent element then the trace pairing $\mathrm{Tr}_{A/k}$ is degenerate.*

Proof. If $x \in A$ is nilpotent then xy is nilpotent for every y , hence has trace 0. Thus $\mathrm{Tr}_{A/k}(xy) = 0$ for all y , i.e. x lies in the left kernel of $\mathrm{Tr}_{A/k}$. $\square$

Definition 2.28 (Discriminant ideal). The *discriminant ideal* $\mathrm{disc}(S/R) \subseteq R$ is the invertible ideal characterized by the local rule: for every nonzero prime $\mathfrak{p} \subset R$ and every $R_{\mathfrak{p}}$ -basis $e_1, \dots, e_n$ of $S_{\mathfrak{p}}$,

$$\mathrm{disc}(S/R)_{\mathfrak{p}} = (\det(\mathrm{Tr}_{L/K}(e_i e_j))) \subseteq R_{\mathfrak{p}}.$$

If R is a PID (for instance $\mathbb{Z}$) then S is a free R -module and we can use Definition 2.8 to compute $\mathrm{disc}(S/R)$. However this is not in general true so we might run into problems. On the other hand the localizations of R are DVRs, so each $S_{\mathfrak{p}}$ is indeed free over $R_{\mathfrak{p}}$ (by the structure theorem for finitely generated modules over PIDs) and we can compute the discriminant locally.

Lemma 2.29. *Suppose that $\mathfrak{p}$ is a prime of R with $\mathfrak{q} \in \mathrm{Spm} \mathfrak{q}$ lying over it. Then the trace pairing controls ramification in the following way.*

$$\begin{array}{llll} \mathfrak{p} \text{ ramifies} & \iff & S/\mathfrak{p} \times S/\mathfrak{p} \rightarrow R/\mathfrak{p} \text{ degenerate} & \iff & S_{\mathfrak{p}} \times S_{\mathfrak{p}} \rightarrow R_{\mathfrak{p}} \text{ imperfect} \\ \mathfrak{q} \text{ ramifies} & \iff & S_{\mathfrak{q}}/\mathfrak{p} \times S_{\mathfrak{q}}/\mathfrak{p} \rightarrow R/\mathfrak{p} \text{ degenerate} & \iff & S_{\mathfrak{q}} \times S_{\mathfrak{q}} \rightarrow R_{\mathfrak{p}} \text{ imperfect} \end{array}$$

Proof. Write $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_g^{e_g}$. Now by Chinese Remainder theorem

$$S/\mathfrak{p}S \cong \bigoplus_{i=1}^g S/\mathfrak{q}_i^{e_i}.$$

The summand $S/\mathfrak{q}_i^{e_i}$ is reduced iff $e_i = 1$, and has nilpotents iff $e_i > 1$. Therefore $S/\mathfrak{p}S$ is reduced iff all $e_i = 1$. Then by Lemma 2.27 ramification makes the reduced trace pairing $\mathrm{Tr}_{(S/\mathfrak{p})/(R/\mathfrak{p})}$ degenerate. In the other direction if $e_i = 1$ then $S/\mathfrak{p}S$ is a direct sum of separable extensions of $R/\mathfrak{p}$ (since $R/\mathfrak{p}$ is assumed perfect). Therefore $\mathrm{Tr}_{(S/\mathfrak{p})/(R/\mathfrak{p})}$ is perfect on each summand so it is perfect on $S/\mathfrak{p}S$.

Taking the localization at some $\mathfrak{q}_k$ isolates a single term of the summand. Our previous argument finishes the theorem for $\mathrm{Tr}_{(S_{\mathfrak{q}}/\mathfrak{p})/(R/\mathfrak{p})}$.

To obtain the second column, note that computing the discriminant commutes with quotient and localization. Then since $R_{\mathfrak{p}}$ is a DVR the discriminant being a non-unit in the localization is equivalent to being 0 in the quotient. $\square$

This gives us a very interesting connection. Prime ramification comes from the failure of the trace pairing to be perfect. Therefore we try to understand when this occurs.

Construction 2.30 (Trace-duality map). The trace pairing on L/K restricts to an R -bilinear pairing on S , and hence an R -linear map

$$\tau_{S/R} : S \longrightarrow S^{\vee} := \mathrm{Hom}_R(S, R), \quad x \longmapsto \mathrm{Tr}(x-).$$

Over K we have shown that the trace duality is an isomorphism $L \xrightarrow{\sim} \mathrm{Hom}_K(L, K)$ because L/K is separable. However over R the pairing may fail to be perfect. In particular $\tau_{S/R}$ is injective and $\mathrm{coker}(\tau_{S/R})$ is a torsion R -module. The *different* is a tool that will allow us to understand "how much" this pairing fails to be perfect over R .

Definition 2.31 (Codifferent / trace dual). The *codifferent* (or *trace dual*) of S/R is the fractional S -ideal

$$S^{\sharp} := \{x \in L : \mathrm{Tr}_{L/K}(xS) \subseteq R\}.$$

as in Construction 2.5.

Remark 2.32. Recall that $S^\sharp$ is identified with $S^\vee$ through the trace-duality map. Then the statement that the pairing is perfect on S is exactly that $S = S^\sharp$ (so that $S \rightarrow S^\vee$ is an isomorphism). Unfortunately the map is not necessarily surjective when restricted to S , so $S^\sharp$ is generally bigger than S . We measure the failure of perfectness by exactly how much bigger it is. However since it's annoying to work with ideals bigger than S , we may as well just invert it.

Definition 2.33 (Different). The *different* of S/R is the inverse fractional ideal

$$\mathcal{D}_{S/R} := (S^\sharp)^{-1} = \{y \in L : yS^\sharp \subseteq S\}.$$

Since $S \subseteq S^\sharp$ (trace of an integral element is integral), we have $\mathcal{D}_{S/R} \subseteq S$.

Now the failure of perfectness is measured by how much smaller than S the different is.

Proposition 2.34. *The discriminant is the ideal norm of the different.*

Proof. We work locally so that R is a DVR. The ideal norm of $S^\sharp$ is by definition the determinant of the map of lattices sending S to $S^\sharp$. Therefore $N(\mathcal{D}_{S/R}) = N(S^\sharp)^{-1}$ is the determinant of the map of lattices sending $S^\sharp$ to S . This is $\text{disc } S/R$ by Lemma 2.4. $\square$

Theorem 2.35. *Let $\mathfrak{p} \subset R$ be a nonzero prime. The following are equivalent*

- (a) $\mathfrak{p} \mid \text{disc}(S/R)$;
- (b) $S_\mathfrak{p}^\sharp \neq S_\mathfrak{p}$ (equivalently, $\mathcal{D}_{S/R, \mathfrak{p}} \neq S_\mathfrak{p}$);
- (c) $\mathfrak{p}$ is ramified.

Moreover if $\mathfrak{q} \in \text{Spm } S$ lies over $\mathfrak{p}$, then the following are equivalent

- (c) $\mathfrak{q} \mid \mathcal{D}_{S/R, \mathfrak{q}}$ (equivalently $S_\mathfrak{q}^\sharp \neq S_\mathfrak{q}$ or $\mathcal{D}_{S/R, \mathfrak{q}} \neq S_\mathfrak{q}$);
- (d) $\mathfrak{q}$ is ramified.

Proof. By Remark 2.32 (b) is the statement that the localized trace pairing $S_\mathfrak{p} \times S_\mathfrak{p} \rightarrow R_\mathfrak{p}$ is imperfect. Moreover (a) is the statement that the reduced trace pairing $S/\mathfrak{p} \times S/\mathfrak{p} \rightarrow R/\mathfrak{p}$ is degenerate. The equivalence (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c) is Lemma 2.29.

By Remark 2.32 (d) is the statement that the localized trace pairing $S_\mathfrak{q} \times S_\mathfrak{q} \rightarrow R_\mathfrak{p}$ is imperfect. Then (c) $\Leftrightarrow$ (d) follows from Lemma 2.29. $\square$

Theorem 2.36 (Monogenic formula for the different). *Suppose $S = R[\alpha]$ for some $\alpha \in S$, and let $f(x) \in R[x]$ be the minimal polynomial of α over K . Then*

$$S^\sharp = \frac{1}{f'(\alpha)} S \quad \text{and hence} \quad \mathcal{D}_{S/R} = (f'(\alpha)) \subseteq S.$$

In particular the discriminant of the power basis satisfies

$$\text{disc}(1, \alpha, \dots, \alpha^{n-1}) = (-1)^{\frac{n(n-1)}{2}} N_{L/K}(f'(\alpha)),$$

where the left-hand side is the determinant $\det(\text{Tr}_{L/K}(\alpha^{i+j-2}))$.

Proof. This is the standard “derivative” description of the trace dual in a monogenic extension: the trace-dual basis to $1, \alpha, \dots, \alpha^{n-1}$ is obtained by dividing appropriate polynomials in α by $f'(\alpha)$, which gives $S^\sharp = (1/f'(\alpha))S$. The discriminant identity then follows by taking norms and comparing with Definition 2.28. $\square$

By Theorem 2.35 a prime $\mathfrak{p} \subset R$ divides $\text{disc}(S/R)$ if and only if some $\mathfrak{q} \mid \mathfrak{p}$ divides $\mathcal{D}_{S/R}$. This says that the different (resp. discriminant) is supported exactly on the ramified primes of S (resp. of R). In particular, only finitely many primes ramify in a number field extension.

3. The Additive Lattice

Note that $\mathbb{Z}$ is the prototypical example of a Dedekind domain. Given a finite separable field extension $L/\mathbb{Q}$ (in this case L is called a *number field*), Theorem 2.12 tells us that the integral closure of $\mathbb{Z}$ in L , called the *ring of integers* and denoted $\mathcal{O}_L$, is always a Dedekind domain.

3.1. The Minkowski embedding

We have become very familiar with the concept that the ring of integers of a PID in a finite separable extension is a lattice. And moreover that the discriminant of the extension is the covolume of the lattice under the trace pairing. In the case of a number field, our intuition for lattices becomes much more precise.

From now on we fix a number field $K/\mathbb{Q}$ of degree n and write $\mathcal{O} := \mathcal{O}_K$. Let r_1 be the number of real embeddings and r_2 the number of *pairs* of complex conjugate embeddings, so that

$$n = r_1 + 2r_2.$$

Choose embeddings

$$\sigma_1, \dots, \sigma_{r_1} : K \hookrightarrow \mathbb{R} \quad \text{and} \quad \tau_1, \dots, \tau_{r_2} : K \hookrightarrow \mathbb{C}$$

where each τ_j is one representative from a conjugate pair (so the full list of embeddings is $\sigma_1, \dots, \sigma_{r_1}, \tau_1, \bar{\tau}_1, \dots, \tau_{r_2}, \bar{\tau}_{r_2}$). We will identify $\mathbb{C} \cong \mathbb{R}^2$ by $x + iy \mapsto (x, y)$.

Construction 3.1 (The additive embedding). Define the map

$$\sigma : K \longrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^n, \quad x \longmapsto (\sigma_1(x), \dots, \sigma_{r_1}(x), \tau_1(x), \dots, \tau_{r_2}(x)),$$

where we view each $\tau_j(x) \in \mathbb{C}$ as a point in $\mathbb{R}^2$ via real and imaginary parts.

Remark 3.2. The minkowski embedding is an injective homomorphism of additive groups, and it extends $\mathbb{R}$ -linearly to an isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$. Furthermore

$$\begin{aligned} N(x) &= \left(\prod_{i=1}^{r_1} \sigma_i(x) \right) \cdot \left(\prod_{i=1}^{r_2} (\operatorname{Re} \tau_i(x))^2 + (\operatorname{Im} \tau_i(x))^2 \right) \\ \operatorname{Tr}(xy) &= \sum_{i=1}^{r_1} \sigma_i(x) \sigma_i(y) + 2 \sum_{i=1}^{r_2} (\operatorname{Re} \tau_i(x) \cdot \operatorname{Re} \tau_i(y) - \operatorname{Im} \tau_i(x) \cdot \operatorname{Im} \tau_i(y)) \end{aligned}$$

which tells us that we can compute the norm and inner product within the embedding. Moreover it is a reasonable norm and inner product to have on $\mathbb{R}^n$.

Proposition 3.3. *The subgroup $\sigma(\mathcal{O}) \subseteq \mathbb{R}^n$ is a lattice. Moreover if $d_K \in \mathbb{Z}$ generates the discriminant ideal $\operatorname{disc}(\mathcal{O}/\mathbb{Z}) \subseteq \mathbb{Z}$, then*

$$\operatorname{covol}(\sigma(\mathcal{O})) = 2^{-r_2} \sqrt{|d_K|}.$$

Proof.

$\sigma(\mathcal{O})$ is a lattice: By Theorem 2.13 (with $R = \mathbb{Z}$) we have $\mathcal{O} \cong \mathbb{Z}^n$ as an abelian group. Choose a $\mathbb{Z}$ -basis $\omega_1, \dots, \omega_n$ of $\mathcal{O}$. Since σ extends to an $\mathbb{R}$ -linear isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^n$, the vectors $\sigma(\omega_1), \dots, \sigma(\omega_n)$ are $\mathbb{R}$ -linearly independent. Therefore

$$\sigma(\mathcal{O}) = \mathbb{Z}\sigma(\omega_1) \oplus \dots \oplus \mathbb{Z}\sigma(\omega_n)$$

is a lattice.

Computing the covolume: Let V be the usual *embedding matrix* (as in Proposition 2.9) for the ordered list of all n embeddings $\sigma_1, \dots, \sigma_{r_1}, \tau_1, \bar{\tau}_1, \dots, \tau_{r_2}, \bar{\tau}_{r_2}$:

$$V_{ij} := \sigma_i(\omega_j) \in \mathbb{C}.$$

By Proposition 2.9 we have

$$d_K = \text{disc}(\omega_1, \dots, \omega_n) = (\det V)^2.$$

Now let M be the real $n \times n$ matrix whose rows are obtained from V by replacing each conjugate pair of rows

$$(\tau_j(\omega_1), \dots, \tau_j(\omega_n)), \quad (\bar{\tau}_j(\omega_1), \dots, \bar{\tau}_j(\omega_n))$$

with the two real rows

$$(\text{Re } \tau_j(\omega_1), \dots, \text{Re } \tau_j(\omega_n)), \quad (\text{Im } \tau_j(\omega_1), \dots, \text{Im } \tau_j(\omega_n)).$$

Concretely, for each j this is a $\mathbb{R}$ -linear change of basis on the 2-dimensional row-span given by

$$\begin{pmatrix} \text{Re } z \\ \text{Im } z \end{pmatrix} \longleftrightarrow \begin{pmatrix} z \\ \bar{z} \end{pmatrix}, \quad \begin{pmatrix} z \\ \bar{z} \end{pmatrix} = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \begin{pmatrix} \text{Re } z \\ \text{Im } z \end{pmatrix}.$$

The determinant of $\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ is $-2i$, so taking all r_2 conjugate pairs we obtain

$$\det V = (\pm(2i)^{r_2}) \det M \quad \Rightarrow \quad |\det M| = 2^{-r_2} |\det V|.$$

Finally the covolume of $\sigma(\mathcal{O})$ is exactly $|\det M|$, so

$$\text{covol}(\sigma(\mathcal{O})) = |\det M| = 2^{-r_2} |\det V| = 2^{-r_2} \sqrt{|(\det V)^2|} = 2^{-r_2} \sqrt{|d_K|}.$$

□

Lemma 3.4. *If $I \subseteq \mathcal{O}$ is a nonzero ideal then $\sigma(I) \subseteq \mathbb{R}^n$ is a sub-lattice of $\sigma(\mathcal{O})$ and*

$$\text{covol}(\sigma(I)) = \text{covol}(\sigma(\mathcal{O})) \cdot [\mathcal{O} : I] = 2^{-r_2} \sqrt{|d_K|} \cdot N(I).$$

Proof. Since I is a $\mathbb{Z}$ -submodule of $\mathcal{O}$ of finite index, its image $\sigma(I)$ is a finite index subgroup of the lattice $\sigma(\mathcal{O})$, hence a sub-lattice. The index is preserved because σ is injective:

$$[\sigma(\mathcal{O}) : \sigma(I)] = [\mathcal{O} : I].$$

Finally for lattices, finite index is the ratio of covolumes:

$$\text{covol}(\sigma(I)) = \text{covol}(\sigma(\mathcal{O})) \cdot [\sigma(\mathcal{O}) : \sigma(I)]$$

and $[\mathcal{O} : I] = N(I)$ by the discussion in the ideal norm section (for $R = \mathbb{Z}$). □

3.2. Minkowski's convex body theorem

Theorem 3.5 (Minkowski). *Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and let $C \subseteq \mathbb{R}^n$ be convex, centrally symmetric ($C = -C$), and measurable. If*

$$\text{vol}(C) > 2^n \text{covol}(\Lambda),$$

then C contains a nonzero lattice point.

Proof. Let $\pi : \mathbb{R}^n \rightarrow \mathbb{R}^n/\Lambda$ be the quotient map. The quotient is a real torus of volume $\text{covol}(\Lambda)$, and π is locally volume preserving. Consider $\frac{1}{2}C = \{x/2 : x \in C\}$. Scaling gives $\text{vol}(\frac{1}{2}C) = \text{vol}(C)/2^n$, so the hypothesis implies

$$\text{vol}\left(\frac{1}{2}C\right) > \text{covol}(\Lambda) = \text{vol}(\mathbb{R}^n/\Lambda).$$

Therefore $\pi(\frac{1}{2}C)$ cannot be injective (pigeonhole for volume), so we can find distinct $x, y \in \frac{1}{2}C$ with $\pi(x) = \pi(y)$. This means $x - y \in \Lambda$ and $x - y \neq 0$.

Finally, since C is convex and symmetric, we have

$$x, y \in \frac{1}{2}C \Rightarrow x - y \in C.$$

Indeed, $x, y \in \frac{1}{2}C$ implies $2x, 2y \in C$, so by symmetry $-2y \in C$ and by convexity

$$(2x) + (-2y) \in C + C \subseteq 2C \Rightarrow x - y \in C.$$

Thus C contains the nonzero lattice point $x - y$. □

3.3. The Minkowski bound

We now feed Theorem 3.5 the specific convex body that remembers the field norm.

Lemma 3.6. *For $T > 0$ define*

$$B_T := \left\{ (x_1, \dots, x_{r_1}, z_1, \dots, z_{r_2}) \in \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} : \sum_{i=1}^{r_1} |x_i| + 2 \sum_{j=1}^{r_2} |z_j| \leq T \right\}.$$

Then B_T is convex, symmetric, and

$$\text{vol}(B_T) = \frac{2^{r_1-r_2}\pi^{r_2}}{n!} T^n.$$

Proof.

- (a) Convexity and symmetry are immediate from the defining inequality.
- (b) For the volume, by symmetry in the real coordinates we may restrict to $x_i \geq 0$ and multiply by 2^{r_1} . For the complex coordinates we use polar coordinates $z_j = \rho_j e^{i\theta_j}$ so $d(\text{Re } z_j) d(\text{Im } z_j) = \rho_j d\rho_j d\theta_j$ and $\int_0^{2\pi} d\theta_j = 2\pi$. Thus

$$\text{vol}(B_T) = 2^{r_1} (2\pi)^{r_2} \int_{\substack{x_i \geq 0, \rho_j \geq 0 \\ \sum x_i + 2 \sum \rho_j \leq T}} \left(\prod_{j=1}^{r_2} \rho_j \right) dx d\rho.$$

Substitute $s_j := 2\rho_j$ so $\rho_j d\rho_j = \frac{1}{4} s_j ds_j$ and hence $\prod \rho_j d\rho_j = 4^{-r_2} \prod s_j ds_j$. We get

$$\text{vol}(B_T) = 2^{r_1} (2\pi)^{r_2} 4^{-r_2} \int_{\substack{x_i \geq 0, s_j \geq 0 \\ \sum x_i + \sum s_j \leq T}} \left(\prod_{j=1}^{r_2} s_j \right) dx ds.$$

Now we integrate out the x_i first. For fixed $s = (s_1, \dots, s_{r_2})$ with $\sum s_j \leq T$, the region in the x -variables is the standard simplex

$$\{x_i \geq 0 : x_1 + \dots + x_{r_1} \leq T - \sum s_j\},$$

which has volume $\frac{(T - \sum s_j)^{r_1}}{r_1!}$. Therefore the integral becomes

$$\int_{\substack{s_j \geq 0 \\ \sum s_j \leq T}} \frac{(T - \sum s_j)^{r_1}}{r_1!} \left(\prod_{j=1}^{r_2} s_j \right) ds.$$

This is a standard simplex integral (one can do it by induction using $\int_0^A (A-t)^a t^b dt = \frac{a!b!}{(a+b+1)!} A^{a+b+1}$), and it evaluates to $\frac{T^n}{n!}$. Putting everything together:

$$\text{vol}(B_T) = 2^{r_1} (2\pi)^{r_2} 4^{-r_2} \cdot \frac{T^n}{n!} = \frac{2^{r_1-r_2}\pi^{r_2}}{n!} T^n.$$

□

Theorem 3.7 (Minkowski lemma). *Let $I \subseteq \mathcal{O}$ be a nonzero ideal. Then there exists $0 \neq \alpha \in I$ such that*

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Proof. We apply Theorem 3.5 to the lattice $\Lambda = \sigma(I) \subseteq \mathbb{R}^n$ and the convex body B_T from Lemma 3.6. By Lemma 3.4 we have

$$\mathrm{covol}(\Lambda) = 2^{-r_2} \sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Using Lemma 3.6, the inequality $\mathrm{vol}(B_T) > 2^n \mathrm{covol}(\Lambda)$ is

$$\frac{2^{r_1-r_2} \pi^{r_2}}{n!} T^n > 2^n \cdot 2^{-r_2} \sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Since $n = r_1 + 2r_2$, this rearranges to

$$T^n > \left(\frac{4}{\pi}\right)^{r_2} n! \sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Choose such a T . Then Minkowski gives a nonzero lattice point $\sigma(\alpha) \in \sigma(I) \cap B_T$, i.e. a nonzero $\alpha \in I$ with

$$\sum_{i=1}^{r_1} |\sigma_i(\alpha)| + 2 \sum_{j=1}^{r_2} |\tau_j(\alpha)| \leq T.$$

Now apply AM–GM to the n nonnegative numbers

$$|\sigma_1(\alpha)|, \dots, |\sigma_{r_1}(\alpha)|, |\tau_1(\alpha)|, |\tau_1(\alpha)|, \dots, |\tau_{r_2}(\alpha)|, |\tau_{r_2}(\alpha)|.$$

Their product is $|\mathrm{N}_{K/\mathbb{Q}}(\alpha)|$ and their average is $\leq T/n$, so

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq \left(\frac{T}{n}\right)^n.$$

Combining with the inequality for T^n gives

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|d_K|} \cdot \mathrm{N}(I),$$

as desired. □

Theorem 3.8 (Minkowski bound). *Every ideal class in Cl_K contains an integral ideal $\mathfrak{a} \subseteq \mathcal{O}$ with*

$$\mathrm{N}(\mathfrak{a}) \leq M_K \quad \text{where} \quad M_K := \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|d_K|}.$$

Proof. Let $c \in \mathrm{Cl}_K$ and choose an integral ideal $I \subseteq \mathcal{O}$ representing c^{-1} . Apply Theorem 3.7 to I to obtain $0 \neq \alpha \in I$ with

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq M_K \cdot \mathrm{N}(I).$$

Since $\alpha \in I$ we have $(\alpha) \subseteq I$, hence

$$\mathfrak{a} := (\alpha)I^{-1} \subseteq \mathcal{O}$$

is an integral ideal. It satisfies $[\mathfrak{a}] = [I^{-1}] = c$, because (α) is principal. Finally by multiplicativity of ideal norm and Remark 2.22 we get

$$\mathrm{N}(\mathfrak{a}) = \mathrm{N}((\alpha)) \mathrm{N}(I^{-1}) = \frac{|\mathrm{N}_{K/\mathbb{Q}}(\alpha)|}{\mathrm{N}(I)} \leq M_K.$$

□

Corollary 3.9. *Every nontrivial extension of $\mathbb{Q}$ is ramified on at least one prime.*

Proof. Since there is at least one ideal class, the Minkowski bound must be larger than 1. This yields

$$\sqrt{|d_K|} \geq \frac{n^n}{n!} \left(\frac{\pi}{4}\right)^{r_2} \geq \frac{n^n}{n!} \left(\frac{\pi}{4}\right)^{n/2}$$

which turns out to be greater than 1 when $n > 1$. $\square$

Remark 3.10. It turns out that if we assume the *Generalized Riemann Hypothesis* then the bound in Theorem 3.8 improves substantially. In this case, we can always find a representative $\mathfrak{a}$ of an ideal class with

$$N(\mathfrak{a}) \leq 12(\ln |d_K|)^2$$

which is much better than $M_K := \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|d_K|}$ for large values of d_K (though for our purposes, the Minkowski bound is often better). In the context of computational algebraic number theory, this is the difference in a polynomial time search and an exponential time search.

3.4. Finiteness of the class number

Now the class number finiteness is essentially a counting statement about sub-lattices of $\mathbb{Z}^n$.

Lemma 3.11. *For every real number $B > 0$ there are only finitely many integral ideals $\mathfrak{a} \subseteq \mathcal{O}$ with $N(\mathfrak{a}) \leq B$.*

Proof. Fix a $\mathbb{Z}$ -basis $\mathcal{O} \cong \mathbb{Z}^n$. If $\mathfrak{a} \subseteq \mathcal{O}$ is an ideal then it is in particular a $\mathbb{Z}$ -submodule, and

$$N(\mathfrak{a}) = [\mathcal{O} : \mathfrak{a}].$$

For a fixed integer $m \geq 1$, any subgroup $H \subseteq \mathbb{Z}^n$ of index m contains $m\mathbb{Z}^n$ by Lemma 2.24. Therefore such H correspond to subgroups of the finite abelian group

$$\mathbb{Z}^n / m\mathbb{Z}^n \cong (\mathbb{Z}/m\mathbb{Z})^n.$$

A finite group has only finitely many subgroups, so there are only finitely many $\mathbb{Z}$ -submodules of index m in $\mathbb{Z}^n$. Finally there are only finitely many integers $m \leq B$, so there are only finitely many ideals with $N(\mathfrak{a}) \leq B$. $\square$

Theorem 3.12. *The class group Cl_K is finite.*

Proof. By Theorem 3.8, every class contains an ideal $\mathfrak{a}$ with $N(\mathfrak{a}) \leq M_K$. By Lemma 3.11 there are only finitely many such ideals. Therefore there are only finitely many ideal classes. $\square$

4. The Units

The other term in the main exact sequence is $\mathcal{O}^\times$. Again the slogan is linear algebra: multiplication becomes addition after taking logs.

Construction 4.1 (The logarithmic embedding). Define

$$\iota : \mathcal{O}^\times \longrightarrow H := \left\{ x \in \mathbb{R}^{r_1+r_2} : \sum_{i=1}^{r_1+r_2} x_i = 0 \right\}$$

by

$$u \longmapsto (\log |\sigma_1(u)|, \dots, \log |\sigma_{r_1}(u)|, 2 \log |\tau_1(u)|, \dots, 2 \log |\tau_{r_2}(u)|).$$

This lands in H because for every unit u ,

$$\sum_{i=1}^{r_1} \log |\sigma_i(u)| + 2 \sum_{j=1}^{r_2} \log |\tau_j(u)| = \log |N_{K/\mathbb{Q}}(u)| = 0.$$

The image is called the *multiplicative lattice*.

Remark 4.2. We have the commutative diagram.

$$\begin{array}{ccccc}
& & \iota & & \\
& \nearrow & & \searrow & \\
K & \xrightarrow{\sigma} & \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2} & \xrightarrow{\log|-|} & \mathbb{R}^{r_1+r_2} \\
\downarrow N_{K/\mathbb{Q}} & & \downarrow N & & \downarrow \text{Tr} \\
\mathbb{Q} & \longrightarrow & \mathbb{R} & \xrightarrow{\log|-|} & \mathbb{R}
\end{array}$$

The middle downward map is given by Remark 3.2 and the right downward map is the standard sum of coordinates on $\mathbb{R}^{r_1+r_2}$. The image of $\mathcal{O}_K$ under σ is the additive lattice and the image of $\mathcal{O}_K^\times$ under ι is the multiplicative lattice. We can consider the image of $\mathcal{O}_K$ under ι , which is some additive subgroup. However its intersection with H corresponds to those elements with norm 1, i.e. the image of $\mathcal{O}_K^\times$ by Theorem 2.18.

Theorem 4.3 (Dirichlet unit theorem). *The kernel of ι is the finite group of roots of unity*

$$\mu_K := \{u \in \mathcal{O}^\times : u^m = 1 \text{ for some } m \geq 1\},$$

and the image $\iota(\mathcal{O}^\times)$ is a lattice in H . In particular

$$\text{rank } \mathcal{O}^\times = r_1 + r_2 - 1.$$

Proof. We separate the argument into three pieces. Extracting the desired properties from the additive lattice.

$\ker \iota = \mu_K$: If $u \in \ker \iota$ then every archimedean absolute value of u is 1. Therefore under the additive embedding from Construction 3.1 we have

$$\sigma(u^m) \in [-1, 1]^n$$

for every $m \geq 1$. Since $\sigma(\mathcal{O}_K)$ is the additive lattice, this box contains only finitely many lattice points. Hence two powers coincide, so u is torsion. The converse is immediate.

$\iota(\mathcal{O}^\times)$ **is discrete**: Fix $B > 0$. If $\|\iota(u)\|_\infty \leq B$ then every embedding of u has absolute value at most e^B . Thus $\sigma(u)$ lies in a bounded box in $\mathbb{R}^n$, and the additive lattice meets a bounded box in only finitely many points. Therefore bounded subsets of H meet $\iota(\mathcal{O}^\times)$ in only finitely many points.

$\iota(\mathcal{O}^\times)$ **has full rank**: Let $\iota(x) = (x_1, \dots, x_{r_1+r_2})$. We would like to construct some unit u with

$$u_k > 0 \quad u_i < 0 \text{ for } i \neq k$$

We construct an infinite sequence of non-zero algebraic integers $\alpha_1, \alpha_2, \dots \in \mathcal{O}_K$ by iteratively applying Minkowski's Convex Body Theorem to a series of boxes. For the first step, choose positive real numbers $c_{1,\tau}$ for each embedding τ such that $\prod_\tau c_{1,\tau}^{N_\tau} = C' > C$ (where $N_\tau = 1$ for real embeddings and 2 for complex embeddings). By Minkowski's Theorem, there exists a non-zero $\alpha_1 \in \mathcal{O}_K$ such that $|\iota_\tau(\alpha_1)| \leq c_{1,\tau}$ for all τ .

For the m -th step (where $m > 1$), choose new positive real bounds $c_{m,\tau}$ such that for all $\tau \neq k$, we strictly shrink the box:

$$c_{m,\tau} < |\iota_\tau(\alpha_{m-1})|$$

To maintain the total volume C' , we compensate by making the bound $c_{m,k}$ sufficiently large. Minkowski's Theorem again guarantees the existence of a non-zero $\alpha_m \in \mathcal{O}_K$ inside this new box, satisfying $|\iota_\tau(\alpha_m)| < c_{m,\tau}$ for all τ . This construction yields an infinite sequence $(\alpha_m)_{m=1}^\infty$ in $\mathcal{O}_K$. By the definition of our bounding boxes, the absolute values in every direction $\tau \neq k$ are strictly decreasing:

$$|\iota_\tau(\alpha_1)| > |\iota_\tau(\alpha_2)| > |\iota_\tau(\alpha_3)| > \dots$$

However, the absolute norm of every α_m is bounded by the constant volume of the box:

$$|N_{K/\mathbb{Q}}(\alpha_m)| = \prod_\tau |\iota_\tau(\alpha_m)|^{N_\tau} \leq C'$$

Since there are only finitely many principal ideals in $\mathcal{O}_K$ with bounded norm there must exist indices $m < n$ such that α_m and α_n generate the exact same principal ideal $(\alpha_m) = (\alpha_n)$. Because they generate the same ideal, they are associates. Thus, there exists a unit $u \in \mathcal{O}_K^*$ such that:

$$\alpha_n = \alpha_m \cdot u \implies u = \frac{\alpha_n}{\alpha_m}$$

We now verify the logarithmic coordinates of u . For any embedding $\tau \neq k$, since $m < n$, our strictly decreasing condition ensures that $|\iota_\tau(\alpha_n)| < |\iota_\tau(\alpha_m)|$. Therefore:

$$|\iota_\tau(u)| = \frac{|\iota_\tau(\alpha_n)|}{|\iota_\tau(\alpha_m)|} < 1$$

Taking the real logarithm, we obtain:

$$\log |\iota_\tau(u)| < 0 \quad \text{for all } \tau \neq k$$

Since u is a unit the sum of all its coordinates is 0. Therefore the remaining k -th term must be strictly positive to balance the equation. This completes the construction of the desired unit. Now construct units $u_1, \dots, u_{r_1+r_2}$ of this form for every index. We claim that they span a subspace of rank at least $r_1 + r_2 - 1$. To see this, assemble them as the rows of a matrix.

$$A_{ij} = \log |\sigma_j(u_i)| \quad A = \begin{bmatrix} + & - & \cdots & - \\ - & + & \cdots & - \\ \vdots & \vdots & \ddots & \vdots \\ - & - & \cdots & + \end{bmatrix}$$

Assume for contradiction that the first $s := r_1 + r_2 - 1$ columns of A are linearly dependent. Then there exist real numbers $c_1, \dots, c_s$, not all zero, such that for every row i :

$$\sum_{j=1}^s A_{ij} c_j = 0$$

Let k be the index of the coefficient with the maximum absolute value, so $|c_k| \geq |c_j|$ for all j , and strictly $|c_k| > 0$. Evaluating the k -th row and isolating the diagonal term yields:

$$\begin{aligned} A_{kk} c_k &= - \sum_{j \neq k}^s A_{kj} c_j \\ A_{kk} |c_k| &\leq \sum_{j \neq k}^s |A_{kj}| |c_j| && \text{triangle inequality} \\ &\leq \sum_{j \neq k}^s (-A_{kj}) |c_k| && \text{maximality and } |A_{kj}| = -A_{kj} \\ A_{kk} &\leq - \sum_{j \neq k}^s A_{kj} \end{aligned}$$

Subtracting the RHS gives $\sum_{j=1}^s A_{kj} \leq 0$. However the sum of the indices along the entire row must be 0, so this sum must be positive. This is a contradiction and we are done.

□

So up to torsion, the unit group is literally a lattice.

Definition 4.4 (Fundamental units). Write $r := r_1 + r_2 - 1$. A collection of units $\epsilon_1, \dots, \epsilon_r \in \mathcal{O}^\times$ is called a system of *fundamental units* if

$$\iota(\mathcal{O}^\times) = \mathbb{Z}\iota(\epsilon_1) \oplus \dots \oplus \mathbb{Z}\iota(\epsilon_r).$$

Corollary 4.5. *If $\epsilon_1, \dots, \epsilon_r$ are fundamental units then every unit $u \in \mathcal{O}^\times$ may be written uniquely as*

$$u = \zeta \epsilon_1^{n_1} \dots \epsilon_r^{n_r} \quad (\zeta \in \mu_K, n_i \in \mathbb{Z}).$$

Equivalently,

$$\mathcal{O}^\times \cong \mu_K \times \mathbb{Z}^r.$$

Proof. Since $\iota(\mathcal{O}^\times)$ is a free abelian group of rank r , choosing a $\mathbb{Z}$ -basis of it and lifting gives the ϵ_i . The only ambiguity in lifting is the kernel of ι , which is exactly μ_K by Theorem 4.3. $\square$

Remark 4.6. This is the multiplicative analogue of the additive lattice. The additive embedding put $\mathcal{O}$ inside $\mathbb{R}^n$ as a lattice of covolume $2^{-r_2} \sqrt{|d_K|}$. The logarithmic embedding puts $\mathcal{O}^\times$ inside the norm-0 hyperplane H as a lattice of rank $r_1 + r_2 - 1$. The covolume of this new lattice is called the *regulator* R_K .

Example 4.7.

- If K is imaginary quadratic then $r_1 = 0$ and $r_2 = 1$, so $r = 0$. Thus $\mathcal{O}^\times = \mu_K$ is finite.
- If K is real quadratic then $r_1 = 2$ and $r_2 = 0$, so $r = 1$. Thus there is a single fundamental unit $\epsilon > 1$ and

$$\mathcal{O}^\times = \{\pm \epsilon^n : n \in \mathbb{Z}\}.$$

So, up to roots of unity, computing the unit group is just finding a basis of the multiplicative lattice.

5. Computing in Number Fields

We've done a lot of theoretical thinking and structural work, however we've not yet figured out how to actually compute stuff. Number theory is one of the most computationally rich fields of mathematics, so it is worth learning that side of it. Therefore in this section we'll learn methods for computing things about number fields.

5.1. Hermite Normal Form

In this section we will learn a tool that allows explicit computation with lattices. While we have previously seen lattices abstractly as finitely generated $\mathbb{Z}$ -modules, to write code that does concrete calculations we need a formalism to represent them.

Proposition 5.1. *The following operations do not change the lattice the columns of a matrix A generates:*

- Swapping two columns.*
- Adding or subtracting a multiple of one column from another.*
- Pick a row r and let c_i, c_j denote the i th and j th columns of A . Using the Euclidean algorithm compute d, x, y so:*

$$d = \gcd(a_{ri}, a_{rj}) \quad xa_{ri} + ya_{rj}$$

then have

$$c_i \leftarrow xc_i + yc_j \quad c_j \leftarrow \frac{a_{rj}}{d}c_i - \frac{a_{ri}}{d}c_j$$

which sends a_{ri} to d , sends a_{rj} to 0, and corresponds to using the invertible operation of the form

$$\begin{pmatrix} x & a_{rj}/d \\ y & -a_{ri}/d \end{pmatrix}$$

on the right.

Definition 5.2 (Hermite Normal Form). Let A be an $n \times k$ integer matrix of rank n , whose columns generate a lattice $\Lambda \subseteq \mathbb{Z}^n$. We call A in *Hermite Normal Form* (HNF) if it is of the form $[H \mid 0]$ (up to column permutation), where H is an $n \times n$ matrix satisfying:

- (a) H is lower triangular ($h_{ij} = 0$ for $i > j$).
- (b) $h_{ii} > 0$ for all i .
- (c) $0 \leq h_{ij} < h_{ii}$ for all $j < i$.

Representing our lattice vectors as columns is standard and algebraically convenient. By enforcing H to be lower triangular, the HNF gives us a canonical choice of basis for any lattice.

Theorem 5.3. *For any integer lattice $\Lambda \in \mathbb{Z}^n$ of rank n , there exists a unique $n \times n$ matrix H in Hermite Normal Form which generates the lattice. Therefore, two matrices generate the same column lattice if and only if they have the same HNF.*

Proof.

Existence: Let A be a matrix whose columns generate the lattice. We apply integer column operations. Working from the top row down, we use Proposition 5.1 (c) on the rows to eliminate entries to the right of the diagonal, making the matrix lower triangular. We then multiply columns by -1 if necessary to ensure $h_{ii} > 0$. Finally, we subtract multiples of column i from column j (for $j < i$) to reduce the off-diagonal entries in the i -th row modulo h_{ii} .

Uniqueness: We proceed by induction on n . Assume uniqueness holds for lattices of rank $n - 1$. We establish the uniqueness of H in three parts:

- *Uniqueness of h_{nn} :* Because H is lower triangular, the intersection of Λ with the n -th coordinate axis, $\Lambda \cap (\{0\}^{n-1} \times \mathbb{Z})$, is a 1-dimensional lattice generated by $(0, \dots, 0, h_{nn})^T$. Since the HNF definition requires $h_{nn} > 0$, h_{nn} is uniquely determined as the positive generator of this sub-lattice.
- *Uniqueness of the top-left block:* Let $\pi(\Lambda) \subset \mathbb{Z}^{n-1}$ be the projection of Λ onto the first $n - 1$ coordinates. The first $n - 1$ columns of H , truncated to their first $n - 1$ rows, form an HNF basis for $\pi(\Lambda)$. By the inductive hypothesis, this $(n - 1) \times (n - 1)$ top-left block is uniquely determined by $\pi(\Lambda)$.
- *Uniqueness of the last column:* Suppose H and H' are two valid HNF bases for Λ . By steps 1 and 2, they share the same h_{nn} and top-left block. Let c_i and c'_i be the i -th columns of H and H' . Since both bases generate Λ , their difference $c_i - c'_i \in \Lambda$. Because their first $n - 1$ coordinates are identical, we have:

$$c_i - c'_i = (0, \dots, 0, h_{ni} - h'_{ni})^T$$

By step 1, this difference must be an integer multiple of $(0, \dots, 0, h_{nn})$. Thus, $h_{ni} \equiv h'_{ni} \pmod{h_{nn}}$. The HNF definition requires that $0 \leq h_{ni}, h'_{ni} < h_{nn}$. The only multiple of h_{nn} that fits the bounded difference is 0. Therefore, $h_{ni} = h'_{ni}$.

Since all components of the matrix are uniquely fixed by the geometry of Λ , the HNF is unique. □

Remark 5.4. Given a $n \times r$ integer matrix A with full rank, we let $\text{HNF}(A)$ denote the unique $n \times n$ Hermite Normal Form matrix generating the same lattice. The above proof (existence) gives an algorithm for computing it.

In algebraic number theory, we often choose a convenient canonical basis, such as a power basis in the next section, and would like to represent our lattice vectors in terms of that. However, the lattice we compute often contains this basis strictly. Therefore to represent our basis vectors we would need to have rational numbers in the matrix. This is a computational disaster, so we use the global denominator.

Definition 5.5 (Global Denominator). Let $\Lambda \subset \mathbb{Q}^n$ be a lattice. We call an integer $d \in \mathbb{Z}_{>0}$ a *global denominator* if $d\Lambda \subseteq \mathbb{Z}^n$.

This lets us represent any lattice basis as $\frac{1}{d}H$, where H is an integer matrix in HNF. Even better, to avoid intermediate coefficient explosion when computing the HNF of a full-rank sub-lattice, we can work entirely modulo a known integer.

Remark 5.6. Let $\Lambda \subseteq \mathbb{Z}^n$ be a full-rank sub-lattice spanned by the columns of an $n \times n$ matrix A and let D be any multiple of the index $[\mathbb{Z}^n : \Lambda]$. We can compute $\text{HNF}(A)$ by working entirely modulo D .

Proof. Note $D\mathbb{Z}^n \subseteq \Lambda$ by Lemma 2.24, the lattice is generated by the columns of A together with the columns of DI_n . Therefore, any column operations used to compute the HNF can be reduced modulo D by adding or subtracting the columns of DI_n . $\square$

Remark 5.7. Writing the basis of a lattice in HNF allows quick computation of how to write any vector in the basis. This is because representing v corresponds to solving

$$Mx = v$$

and we can determine x_i inductively from the i th row, starting from the top, since at each step the equation only has one unknown.

The Hermite Normal Form is the standard way to crush a bunch of generators into a nice basis, and moreover it is very good for many computations. Therefore in the next sections, when I say choose a basis, unless otherwise implied, it is understood that the basis is the Hermite Normal basis.

5.2. Computing the Discriminant

In general, the first step to understanding a number field is trying to learn something about its discriminant.

Example 5.8 (The Discriminant of a Polynomial and its Field). Let $f(x) \in R[x]$ be a monic irreducible polynomial of degree n with roots $\alpha_1, \dots, \alpha_n$ in a splitting field L over $K = \text{Frac}(R)$. There are three ways we may define Δ_f (the discriminant of f):

Root Differences: Defined essentially as the product of squared differences of the roots. This vanishes if and only if roots collide (i.e., f is inseparable).

$$\Delta_f = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2$$

The Resultant: Calculable without knowing the roots, using the resultant of f and its derivative f' :

$$\Delta_f = (-1)^{\frac{n(n-1)}{2}} \text{Res}(f, f')$$

We still need to define the *resultant* that we have used in the formula.

Let $f(x) = a_n x^n + \dots + a_0$ and $g(x) = b_m x^m + \dots + b_0$ be polynomials in $F[x]$ of degrees n and m respectively. The *Sylvester matrix* of f and g is the $(n+m) \times (n+m)$ matrix:

$$S_{f,g} = \begin{pmatrix} a_n & a_{n-1} & \dots & a_0 & 0 & \dots & 0 \\ 0 & a_n & \dots & a_1 & a_0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & a_n & \dots & \dots & a_0 \\ b_m & b_{m-1} & \dots & b_0 & 0 & \dots & 0 \\ 0 & b_m & \dots & b_1 & b_0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & b_m & \dots & \dots & b_0 \end{pmatrix}.$$

Now $\text{Res}(f, g)$, is defined as the determinant of $S_{f,g}$, or alternatively as

$$\text{Res}(f, g) = a_0^e b_0^d \prod_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} (\lambda_i - \mu_j)$$

where λ_i and μ_j are the roots of f and g respectively.

Derivative Values: Expresses Δ_f via the values of the derivative at the roots.

$$\Delta_f = (-1)^{\frac{n(n-1)}{2}} \prod_{i=1}^n f'(\alpha_i) = (-1)^{\frac{n(n-1)}{2}} n^n \prod_{i=1}^{n-1} f(\beta_i)$$

where the β_i are the roots of f' .

Proof. These equivalences may be found in Neukirch. Intuitively, they all are different ways you might try to test whether f and f' share roots.

- The product $(\alpha_i - \alpha_j)^2$ computes whether f has a double root, which is true if and only if f and f' share a root.
- If f and f' share roots then by Bezout's theorem there are polynomials a and b (found by Euclid's algorithm) with:

$$af + bf' = 0 \quad \deg a < \deg f' \quad \deg b < \deg f$$

This forces the determinant of the Sylvester matrix to be 0 since it corresponds exactly to a linear dependence among the rows.

- If f and f' share roots then evaluating one at the roots of the other will give 0.

□

Theorem 5.9. *If f is an irreducible polynomial and α is one of its roots then $\text{disc } R[\alpha]/R$ (with respect to the trace pairing on $K(\alpha)/K$) is exactly Δ_f .*

Proof. Since f is irreducible it is the minimal polynomial of α . We may use the power basis $\{1, \alpha, \dots, \alpha^{n-1}\}$ to compute $\text{disc } R[\alpha]/R$. By Proposition 2.9 this is the square of the determinant of the embedding matrix $V_{ij} = \sigma_i(\alpha^{j-1})$. This matrix is precisely the Vandermonde matrix of the roots $\alpha_1, \dots, \alpha_n$ of f . Thus:

$$\Delta(1, \dots, \alpha^{n-1}) = (\det V)^2 = \prod_{i < j} (\alpha_i - \alpha_j)^2 = \Delta_f.$$

□

This tells us that if $\mathcal{O}_L/\mathcal{O}_K$ is *monogenic* (so $\mathcal{O}_L = \mathcal{O}_K[\alpha]$), then the discriminant $\Delta_{L/K}$ is exactly Δ_f where f is the minimal polynomial of α .

Algorithm 5.10. Computing huge resultants is generally hard, however there is an algorithm to do it quickly in a Euclidean domain. We will mainly focus on the Sylvester matrix definition.

- Use the Hadamard bound to see that the resultant's value is bounded by the product of the Euclidean norms of the rows.
- Pick small primes p_i which multiply to a number bigger than the bound in (a).
- Use (d) to compute the resultant mod p_i by working in R/p_i .
- Note the formula

$$\text{Res}(f, g) = (-1)^{mn} \text{lc}(g)^{m - \deg h} \cdot \text{Res}(h, g)$$

where lc is the leading coefficient of g and h is the remainder of f after division by g . This lets us compute using the Euclidean algorithm, which is not very expensive in finite fields.

- Use the Chinese Remainder Theorem to find the actual resultant.

For a human, you should probably ignore the parts for working in mod p and just use the Euclidean algorithm.

Proposition 5.11 (Discriminant and index). *If $A \subseteq B$ are orders in K , then*

$$\text{disc}(A/\mathbb{Z}) = [B : A]^2 \text{disc}(B/\mathbb{Z}).$$

In particular,

$$\text{disc}(A/\mathbb{Z}) = [\mathcal{O} : A]^2 \text{disc}(\mathcal{O}/\mathbb{Z}).$$

Proof. Recall that changing basis scales the discriminant by the square of the change of basis matrix. $\square$

Algorithm 5.12. This gives the following general algorithm to compute the discriminant:

- (a) Find a primitive element α of the field extension. The discriminant of the power basis is equal to Δ_f where f is the minimal polynomial of f .
- (b) While computing the ring of integers, we enlarge this order in iterative steps. At each step, use Proposition 5.11 to keep track of the running discriminant.

Now we will learn a shortcut which is often helpful when one can break a field extension into parts.

Theorem 5.13 (Transitivity of the Different). *Let $F \subset K \subset L$ be a tower of finite separable extensions and let $\mathcal{O}_F \subset \mathcal{O}_K \subset \mathcal{O}_L$ be their respective rings of integers. Then the relative different ideals satisfy the transitivity relation:*

$$\mathcal{D}_{L/F} = \mathcal{D}_{L/K} \cdot \mathcal{D}_{K/F}$$

where $\mathcal{D}_{K/F}$ is viewed as the ideal it generates in $\mathcal{O}_L$.

Proof. It is much easier to prove the corresponding identity for the inverse different, also known as the codifferent. Recall that for a finite separable extension E/k , the codifferent $\mathcal{D}_{E/k}^{-1}$ is defined as the fractional ideal:

$$\mathcal{D}_{E/k}^{-1} = \{x \in E \mid \text{Tr}_{E/k}(x\mathcal{O}_E) \subseteq \mathcal{O}_k\}$$

We want to show that $\mathcal{D}_{L/F}^{-1} = \mathcal{D}_{L/K}^{-1} \cdot \mathcal{D}_{K/F}^{-1}$.

Let $x \in L$. By the definition of the codifferent for the extension L/F , we have:

$$\begin{aligned} x \in \mathcal{D}_{L/F}^{-1} &\iff \text{Tr}_{L/F}(x\mathcal{O}_L) \subseteq \mathcal{O}_F \\ &\iff \text{Tr}_{K/F}(\text{Tr}_{L/K}(x\mathcal{O}_L)) \subseteq \mathcal{O}_F && \text{transitivity of trace} \\ &\iff \text{Tr}_{K/F}(\text{Tr}_{L/K}(x\mathcal{O}_L)\mathcal{O}_K) \subseteq \mathcal{O}_F && \text{linearity of trace on base field} \\ &\iff \text{Tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{D}_{K/F}^{-1} && \text{definition of } \mathcal{D}_{K/F}^{-1} \\ &\iff \text{Tr}_{L/K}(x\mathcal{O}_L\mathcal{D}_{K/F}) \subseteq \mathcal{O}_F && \text{linearity of trace on base field} \\ &\iff x\mathcal{D}_{K/F} \subseteq \mathcal{D}_{L/K}^{-1} && \text{definition of } \mathcal{D}_{L/K}^{-1} \\ &\iff x \in \mathcal{D}_{L/K}^{-1} \cdot \mathcal{D}_{K/F}^{-1} \end{aligned}$$

Taking the inverse of both sides immediately yields the transitivity formula for the different:

$$\mathcal{D}_{L/F} = \mathcal{D}_{L/K} \cdot \mathcal{D}_{K/F}$$

$\square$

Theorem 5.14 (Tower Law for Discriminants). *For a tower of finite separable extensions $L/K/F$:*

$$\Delta_{L/F} = N_{K/F}(\Delta_{L/K}) \cdot \Delta_{K/F}^{[L:K]}$$

Proof. Recall the transitivity of the relative different ideal:

$$\mathcal{D}_{L/F} = \mathcal{D}_{L/K} \cdot \mathcal{D}_{K/F}$$

The discriminant the norm of the different, $\Delta_{L/F} = N_{L/F}(\mathcal{D}_{L/F})$. Applying $N_{L/F}$ to the transitivity relation and using the transitivity of the norm ($N_{L/F} = N_{K/F} \circ N_{L/K}$), we have:

$$\begin{aligned}\Delta_{L/F} &= N_{L/F}(\mathcal{D}_{L/K}) \cdot N_{L/F}(\mathcal{D}_{K/F}) \\ &= N_{K/F}(N_{L/K}(\mathcal{D}_{L/K})) \cdot N_{K/F}(N_{L/K}(\mathcal{D}_{K/F}))\end{aligned}$$

By definition, $N_{L/K}(\mathcal{D}_{L/K}) = \Delta_{L/K}$. Furthermore, since $\mathcal{D}_{K/F}$ is an ideal of $\mathcal{O}_K$, its norm from L to K is simply the ideal raised to the degree of the extension: $N_{L/K}(\mathcal{D}_{K/F}) = \mathcal{D}_{K/F}^{[L:K]}$. Substituting these in:

$$\Delta_{L/F} = N_{K/F}(\Delta_{L/K}) \cdot N_{K/F}(\mathcal{D}_{K/F})^{[L:K]}$$

Recognizing $N_{K/F}(\mathcal{D}_{K/F})$ as $\Delta_{K/F}$ completes the argument:

$$\Delta_{L/F} = N_{K/F}(\Delta_{L/K}) \cdot \Delta_{K/F}^{[L:K]}$$

□

Corollary 5.15. *If two number fields have coprime discriminant then they intersect trivially.*

Proof. By the tower law for discriminant, every prime dividing $d_{K \cap L}$ must divide d_K and d_L . Therefore $\Delta_{K \cap L} = 1$ by the hypothesis. Now Corollary 3.9 yields the claim. □

Theorem 5.16. *Let K and L be number fields with rings of integers $\mathcal{O}_K$ and $\mathcal{O}_L$, and discriminants d_K and d_L . If $\gcd(d_K, d_L) = 1$, then*

$$\Delta_{KL} = d_K^{[L:\mathbb{Q}]} \cdot d_L^{[K:\mathbb{Q}]}$$

Proof. By Theorem 5.29 we have $\mathcal{O}_{KL} = \mathcal{O}_K \cdot \mathcal{O}_L$. Moreover by Corollary 5.15 they intersect trivially so $\mathcal{O}_{KL} \cong \mathcal{O}_K \otimes \mathcal{O}_L$ and the trace pairing is exactly the one inherited on the tensor product. This gives the result. □

5.3. Zassenhaus Round 2

In this section I will give the machinery necessary to perform the "Round 2" algorithm for finding the ring of integers. The nice thing about this is that it is a relatively practical algorithm for humans to use, and also has pretty solid asymptotics for computers (better algorithms do exist).

The computation of $\mathcal{O}$ is local at the primes dividing the index of a convenient starting order. In practice one usually begins with a monogenic order

$$A = \mathbb{Z}[\alpha] \subseteq \mathcal{O}$$

coming from an integral primitive element α . There are then two tools:

- Dedekind's criterion, which often proves immediately that a given prime does *not* divide $[\mathcal{O} : A]$;
- the *Round 2* step, which enlarges A at the remaining bad primes until it becomes maximal.

Intuitively, if we don't have enough integers, our discriminant will be too large since our lattice is too small. This will give "fake ramification", which we can check on each prime.

Corollary 5.17 (The first discriminant test). *If a prime p divides $[\mathcal{O} : A]$, then p^2 divides $\text{disc}(A/\mathbb{Z})$.*

Proof. This is immediate from Proposition 5.11. □

So only the primes p with $p^2 \mid \text{disc}(A/\mathbb{Z})$ can possibly be bad.

Proposition 5.18 (Local description of p -maximality). *Let $A \subseteq \mathcal{O}$ be an order and let p be a rational prime. We say that A is p -maximal if it satisfies one of the equivalent conditions:*

- (a) $p \nmid [\mathcal{O} : A]$;

(b) for every prime $\mathfrak{p} \subseteq A$ with $\mathfrak{p} \cap \mathbb{Z} = (p)$, one has

$$A_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}};$$

(c) for every such $\mathfrak{p}$, the local ring $A_{\mathfrak{p}}$ is a DVR.

Proof. Let $M := \mathcal{O}/A$, a finite abelian group of order $[\mathcal{O} : A]$.

(a) $\Leftrightarrow$ (b): The condition $p \nmid [\mathcal{O} : A]$ is equivalent to saying that the p -primary subgroup of M is zero. If $\mathfrak{p} \cap \mathbb{Z} = (p)$, then every prime $q \neq p$ is a unit in $A_{\mathfrak{p}}$, so localizing at $\mathfrak{p}$ kills all q -primary torsion. Thus $M_{\mathfrak{p}} = 0$ for every $\mathfrak{p} \mid p$ if and only if the p -primary subgroup of M is zero. But

$$M_{\mathfrak{p}} \cong \mathcal{O}_{\mathfrak{p}}/A_{\mathfrak{p}},$$

so $M_{\mathfrak{p}} = 0$ is exactly the equality $A_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$.

(b) $\Rightarrow$ (c): Each $\mathcal{O}_{\mathfrak{p}}$ is a localization of the Dedekind domain $\mathcal{O}$, hence a DVR.

(c) $\Rightarrow$ (b): The integral closure of $A_{\mathfrak{p}}$ in K is $\mathcal{O}_{\mathfrak{p}}$. If $A_{\mathfrak{p}}$ is a DVR, then it is integrally closed, so it already equals its integral closure. □

Corollary 5.19. $A = \mathcal{O}$ if and only if A is p -maximal for all primes p .

Proof. We can check whether two rings are equal locally. □

Lemma 5.20 (The local idealizer test). *Let R be a one-dimensional Noetherian local domain with maximal ideal $\mathfrak{m}$. Then*

$$(\mathfrak{m} : \mathfrak{m}) = R \quad \Longleftrightarrow \quad R \text{ is a DVR.}$$

Proof. If R is a DVR and $\mathfrak{m} = (\pi)$, then

$$(\mathfrak{m} : \mathfrak{m}) = (\pi R : \pi R) = (R : R) = R$$

because R is an integral domain.

Conversely, suppose R is not a DVR. Then $\mathfrak{m}$ must not be principle and is therefore not invertible by Theorem 1.4. In this case we must have $\mathfrak{m}\mathfrak{m}^{-1} = \mathfrak{m}$ (it contains $\mathfrak{m}$ and is contained in R) so it suffices to show $R \not\subseteq \mathfrak{m}^{-1}$. This is due to Lemma 1.5. □

Theorem 5.21 (One step of Round 2). *Let $A \subseteq K$ be an order, let p be a rational prime, and set:*

$$I := \sqrt{pA} \quad A' := (I : I)$$

Then A' an order satisfying:

$$A \subseteq A' \subseteq \frac{1}{p}A$$

Moreover $[A' : A]$ is a power of p , and

$$A' = A \quad \Longleftrightarrow \quad A \text{ is } p\text{-maximal.}$$

Proof. Note $pA \subseteq I \subseteq A$ so

$$x \in (I : I) \implies x \in (A : pA) = \frac{1}{p}A$$

and $(I : I) \subseteq \frac{1}{p}A$. Thus $(I : I)$ is a finitely generated torsion-free $\mathbb{Z}$ -module of full rank, i.e. an order. Now $[A' : A]$ must divide $[\frac{1}{p}A : A]$ and is therefore a power of p . Finally, let $\mathfrak{p} \subseteq A$ be a prime with $\mathfrak{p} \cap \mathbb{Z} = (p)$. Localizing gives

$$I_{\mathfrak{p}} = \sqrt{pA_{\mathfrak{p}}} = \mathfrak{p}A_{\mathfrak{p}},$$

because $A_{\mathfrak{p}}$ is local and $\mathfrak{p}A_{\mathfrak{p}}$ is its unique prime above p . Hence

$$A'_{\mathfrak{p}} = (I_{\mathfrak{p}} : I_{\mathfrak{p}}) = (\mathfrak{p}A_{\mathfrak{p}} : \mathfrak{p}A_{\mathfrak{p}}).$$

By Lemma 5.20, this equals $A_{\mathfrak{p}}$ if and only if $A_{\mathfrak{p}}$ is a DVR. By Proposition 5.18, the latter is equivalent to A being p -maximal. □

This is very mathematically elegant and gives a good way of computing the ring of integers.

Remark 5.22. The idealizer is computed by linear algebra over $\mathbb{F}_p$. The first step is to compute $I = \sqrt{pA}$. Since $pA \subseteq I$ we work entirely in A/pA .

- (a) Pick a basis $\beta_1, \dots, \beta_n$ for A . Compute the $\mathbb{F}_p$ matrix M^r corresponding to the action of each β_r :

$$\beta_r \beta_j \equiv \sum_i M_{ij}^r \beta_i \pmod{p}$$

- (b) Using (a) compute $a_{ij} \in \mathbb{F}_p$ with:

$$\beta_j^p \equiv \sum_i a_{ij} \beta_i \pmod{p}$$

and note by Freshman's dream the p th power map is exactly the matrix P with coefficients a_{ij} .

- (c) Choose m minimal so $p^m > n$. By Freshman's dream the p^m power map is exactly P^m . Now I corresponds exactly to the null space of this matrix. Therefore compute a β_i -basis $\gamma_1, \dots, \gamma_n$ for I using RREF.

Now we will compute the full idealizer $A' = (I : I)$. We use the fact that $A \subseteq A' \subseteq \frac{1}{p}A$ to work in $(\frac{1}{p}A)/A \cong A/pA$.

- (d) Compute matrices N^r corresponding to the action of β_i on I :

$$\beta_r \gamma_j \equiv \sum_i N_{ij}^r \gamma_i \pmod{p} \quad N^r = B M^r B^{-1}$$

Where B is the change of basis matrix from β_i to γ_i that we computed in (c).

- (e) For $x = \frac{1}{p}(c_1\beta_1 + \dots + c_n\beta_n)$ we have

$$x \in (I : I) \iff c_1 N^1 + \dots + c_n N^n \equiv 0 \pmod{p}$$

which is a system of n^2 congruences modulo p . We compute a basis for the idealizer by taking RREF of a $n^2 \times n$ matrix.

Algorithm 5.23 (Round 2 algorithm). Start with any order $A_0 \subseteq \mathcal{O}$. For each prime p with $p^2 \mid \text{disc}(A_0/\mathbb{Z})$, repeat the following step (using Remark 5.22):

$$A \leftarrow (\sqrt{pA} : \sqrt{pA})$$

until it stabilizes. After doing this for every such prime p , the resulting order is $\mathcal{O}$.

Proof. By Theorem 5.21, the iteration at a fixed prime p stabilizes exactly when the current order is p -maximal. Each strict enlargement

$$A \subsetneq A' := (\sqrt{pA} : \sqrt{pA})$$

has index a positive power of p . Therefore Proposition 5.11 shows that

$$v_p(\text{disc}(A'/\mathbb{Z})) < v_p(\text{disc}(A/\mathbb{Z})).$$

Since this valuation is a nonnegative integer, the iteration must terminate.

When the process is finished for every prime p with $p^2 \mid \text{disc}(A_0/\mathbb{Z})$, the final order is p -maximal for every prime p . Hence the final order is $\mathcal{O}$ itself. $\square$

The last bit of machinery is a special case, it turns out that when the order is monogenic the computation is much easier.

Theorem 5.24 (Dedekind's Criterion). *Let $K = \mathbb{Q}(\alpha)$ be a number field where α is an algebraic integer with minimal monic polynomial $f(x) \in \mathbb{Z}[x]$ of degree n . Let $A = \mathbb{Z}[\alpha]$ and p be a prime. Let the factorization of $f(x)$ modulo p be $\bar{f}(x) = \prod_i \bar{g}_i(x)^{e_i}$.*

Let $g_i(x)$ be monic lifts of the $\bar{g}_i(x)$ and define the following polynomials in $\mathbb{Z}[x]$:

- $h(x) = \prod_i g_i(x)^{e_i - 1}$
- $g(x) = \prod_i g_i(x)$
- $k(x) = \frac{f(x) - g(x)h(x)}{p}$

Then

- The order A is p -maximal if and only if $\gcd(\bar{h}(x), \bar{k}(x)) = 1$ in $\mathbb{F}_p[x]$.
- If A is not p -maximal, let $d(x) = \gcd(\bar{g}(x), \bar{h}(x), \bar{k}(x))$ in $\mathbb{F}_p[x]$ and let $U(x) \in \mathbb{Z}[x]$ be any monic lift of $\frac{\bar{f}(x)}{d(x)}$. A strictly larger order is given by

$$A' = (\sqrt{pA} : \sqrt{pA}) = A \left[\frac{U(\alpha)}{p} \right]$$

- A' has basis

$$1, \alpha, \alpha^2, \dots, \alpha^{\deg U - 1}, \frac{U(\alpha)}{p}, \frac{\alpha U(\alpha)}{p}, \dots, \frac{\alpha^{\deg(d)-1} U(\alpha)}{p}$$

so the index $[A' : A]$ is $p^{\deg(d)}$.

Proof. We utilize the Chinese Remainder Theorem:

$$\frac{A}{pA} \cong \frac{\mathbb{F}_p[x]}{f(x)} \cong \prod_i \frac{\mathbb{F}_p[x]}{\bar{g}_i(x)^{e_i}}$$

Note the p -radical ideal is $I = \sqrt{pA} = pA + g(\alpha)A$. We will compute the multiplier ring $(I : I) = \{x \in K \mid xI \subseteq I\}$. Suppose $x \in (I : I)$. By Theorem 5.21 we can let $x = v(\alpha)/p$ for $v \in \mathbb{Z}[x]$. Now

$$x \in (A : I) \implies \bar{v}(\alpha)\bar{g}(\alpha) = 0 \pmod{p}$$

So we can write $x = \frac{m(\alpha)h(\alpha)}{p}$ for $m(x) \in \mathbb{Z}[x]$. For $x \in (I : I)$, we require:

- $x \cdot p \in I \implies \bar{g} \mid \bar{m}\bar{h}$ in $\mathbb{F}_p[x]$.
- $x \cdot g(\alpha) \in I \implies \frac{m(\alpha)h(\alpha)g(\alpha)}{p} \equiv -m(\alpha)k(\alpha) \in I \implies \bar{g} \mid \bar{m}\bar{k}$ in $\mathbb{F}_p[x]$.

These hold simultaneously if and only if $\bar{g}$ divides $\bar{m} \cdot \gcd(\bar{h}, \bar{k})$. Let $d = \gcd(\bar{g}, \bar{h}, \bar{k})$. The minimal degree multiplier is $\bar{m} = \frac{\bar{g}}{d}$. Therefore it is necessary and sufficient that modulo p , x is a multiple of

$$\frac{(\bar{g}/d)h(\alpha)}{p} = \frac{(\bar{f}/d)}{p} \equiv \frac{U(\alpha)}{p} \pmod{A}$$

So adjoining $\frac{U(\alpha)}{p}$ yields the overorder A' . Moreover $A' = A$ if and only if:

$$U(\alpha) = 0 \pmod{p} \iff d = 1$$

To compute the index $[A' : A]$, consider the standard $\mathbb{Z}$ -basis of A : $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$. Since $U(x)$ is monic of degree $n - \deg(d)$ and $\bar{U}(x)$ is a factor of $\bar{f}(x)$, the elements $\frac{U(\alpha)}{p}, \frac{\alpha U(\alpha)}{p}, \dots, \frac{\alpha^{\deg(d)-1} U(\alpha)}{p}$ can replace the basis elements $\alpha^{n-\deg(d)}, \dots, \alpha^{n-1}$ to form a $\mathbb{Z}$ -basis for A' .

The transition matrix to this new basis is upper-triangular. Its diagonal consists of $n - \deg(d)$ entries equal to 1 (for the unchanged lower-degree basis elements) and $\deg(d)$ entries equal to $\frac{1}{p}$ (for the new elements). The determinant of this transformation matrix is $p^{-\deg(d)}$, proving that the index $[A' : A]$ is exactly $p^{\deg(d)}$. $\square$

Since computing gcd or dividing polynomials in finite fields is very cheap, Dedekind's criterion is preferred when the order is monogenic. When the order isn't, computing round 2 on bad primes requires a lot of linear algebra.

Remark 5.25. A natural way to remember the Dedekind criterion is the following. Keeping the notation of Theorem 5.24:

- Consider only the $\bar{g}_i$ with $e_i \geq 2$. We say each such $\bar{g}_i$ passes the criterion if $\bar{g}_i \nmid \bar{k}$.
- If every such $\bar{g}_i$ passes, then A is p -maximal.
- Otherwise, let d be the product of the $\bar{g}_i$ which fail and let U be a monic lift of $\bar{f}/d$. Then

$$A' = (I : I) = A + A \frac{U(\alpha)}{p}$$

is a bigger order with index $[A' : A]$ equal to $p^{\deg d}$.

Algorithm 5.26. To compute with Dedekind criterion, it is not actually necessary to factor f since our computations are not sensitive to which monic lifts we choose. Instead we can cheat with some clever uses of the gcd (Computed fast via Euclidean algorithm over $\mathbb{F}_p$).

(a) Compute:

- $\bar{h}(x) = \gcd(\bar{f}(x), \bar{f}'(x))$
- $\bar{g}(x) = \bar{f}(x)/\bar{h}(x)$
- $k(x) = \frac{f(x) - g(x)h(x)}{p}$ for any monic lifts $g(x), h(x)$ of $\bar{g}(x), \bar{h}(x)$.

(b) Compute $\gcd(\bar{f}(x), \bar{h}(x))$, if it is 1 we are done.

(c) Otherwise compute $d(x) = \gcd(\bar{f}(x), \bar{h}(x), \bar{k}(x))$ by computing the gcd of $\bar{k}(x)$ with the result of (b). Note that the index will be $p^{\deg d}$.

(d) Let $U(x)$ be any monic lift of $\bar{f}(x)/d(x)$ and have

$$A' = A \left[\frac{U(x)}{p} \right]$$

which has the $\mathbb{Z}$ -basis:

$$1, \alpha, \alpha^2, \dots, \alpha^{\deg U - 1}, \frac{U(\alpha)}{p}, \frac{\alpha U(\alpha)}{p}, \dots, \frac{\alpha^{\deg(d)-1} U(\alpha)}{p}$$

Algorithm 5.27. In practice one usually proceeds as follows.

- Start with a convenient monogenic order $A = \mathbb{Z}[\alpha]$. The power basis is the canonical basis we do our math in (so we finish with basis elements expressed as polynomials in α).
- Compute $\text{disc}(f) = \text{disc}(A/\mathbb{Z})$ using Algorithm 5.10. By Corollary 5.17, only the primes p with $p^2 \mid \text{disc}(f)$ can be bad.
- For each such p , use Dedekind's criterion (Algorithm 5.26). If it fails, let A'_p be the new overorder.
- Send:

$$A \leftarrow A'_{p_1} + \dots + A'_{p_r}$$

which is computed via HNF by adjoining all the $\mathbb{Z}$ -basis elements we got in Algorithm 5.26. The new discriminant is the old divided by the product of the squares of the indices we got.

- For every prime for which Dedekind's criterion failed, and whose square divides the new discriminant, perform the round 2 algorithm (Algorithm 5.23) to expand A .

The round 2 is quite expensive, which is why this algorithm is almost designed to avoid it for as long as possible.

Problem 5.28. Let

$$K = \mathbb{Q}(\sqrt{5}), \quad A = \mathbb{Z}[\sqrt{5}], \quad f(x) = x^2 - 5.$$

and perform this algorithm to obtain the ring of integers.

The algorithm is extreme overkill in the above case, however it is the first nontrivial quadratic example. Now that we have a general method, I'll list out some shortcuts that are often useful.

Theorem 5.29. *Let K and L be number fields with rings of integers $\mathcal{O}_K$ and $\mathcal{O}_L$, and discriminants d_K and d_L . If $\gcd(d_K, d_L) = 1$, then the ring of integers of the compositum KL is $\mathcal{O}_{KL} = \mathcal{O}_K \mathcal{O}_L$.*

Proof. The inclusion $\mathcal{O}_K \mathcal{O}_L \subseteq \mathcal{O}_{KL}$ holds trivially. To prove the reverse inclusion, let $\{\alpha_1, \dots, \alpha_n\}$ be an integral basis for $\mathcal{O}_K$. Because $\gcd(d_K, d_L) = 1$, the fields K and L are linearly disjoint over $\mathbb{Q}$, which implies that $\{\alpha_1, \dots, \alpha_n\}$ is also a basis for KL over L . Let $x \in \mathcal{O}_{KL}$ be an arbitrary algebraic integer in the compositum. We can express it uniquely as $x = \sum_{i=1}^n c_i \alpha_i$ for some coefficients $c_i \in L$. We will show that $c_i \in \mathcal{O}_L$ using the trace pairing:

- Multiplying x by α_j and taking the relative trace down to L yields $\text{Tr}_{KL/L}(x\alpha_j) = \sum_{i=1}^n c_i \text{Tr}_{KL/L}(\alpha_i \alpha_j)$.
- Since x and α_j are both algebraic integers, their product $x\alpha_j$ is an algebraic integer. The trace of an algebraic integer is an algebraic integer, so $\text{Tr}_{KL/L}(x\alpha_j) \in \mathcal{O}_L$.
- Furthermore, because the α_i reside in K , the relative trace of their products restricts to the absolute trace: $\text{Tr}_{KL/L}(\alpha_i \alpha_j) = \text{Tr}_{K/\mathbb{Q}}(\alpha_i \alpha_j) \in \mathbb{Z}$.
- We now have a linear system of equations $M\vec{c} = \vec{t}$, where M is an $n \times n$ matrix with integer entries $M_{j,i} = \text{Tr}_{K/\mathbb{Q}}(\alpha_i \alpha_j)$, $\vec{c}$ is the column vector of the unknown c_i , and $\vec{t}$ is a column vector with entries in $\mathcal{O}_L$.
- By Cramer's rule, we can solve for the coefficients as $c_i = \frac{\det(M_i)}{\det(M)}$, where M_i is the matrix formed by replacing the i -th column of M with $\vec{t}$.
- The determinant of M is exactly the discriminant d_K . Since M_i has entries entirely in $\mathcal{O}_L$, its determinant must also lie in $\mathcal{O}_L$. Therefore, $d_K c_i \in \mathcal{O}_L$.

Because $d_K c_i \in \mathcal{O}_L$ for all i we have $d_K x \in \mathcal{O}_K \mathcal{O}_L$. By a symmetric argument, we obtain $d_L x \in \mathcal{O}_K \mathcal{O}_L$. Since $\gcd(d_K, d_L) = 1$, by Bézout's identity there exist integers $u, v \in \mathbb{Z}$ such that $ud_K + vd_L = 1$. We can therefore write x as:

$$x = 1 \cdot x = (ud_K + vd_L)x = u(d_K x) + v(d_L x)$$

Since both $d_K x$ and $d_L x$ belong to the ring $\mathcal{O}_K \mathcal{O}_L$, it follows that $x \in \mathcal{O}_K \mathcal{O}_L$. Thus, $\mathcal{O}_{KL} \subseteq \mathcal{O}_K \mathcal{O}_L$, completing the proof. $\square$

5.4. Computing Prime Behaviour

In this section I will discuss two tools for computing how primes behave. That is the Dedekind-Kummer theorem that we have seen, as well as Newton Polygons for when this method fails.

5.5. Computing the Class Group

5.6. Pell Convergents

Now we will learn how to compute the units of quadratic fields. At a conceptual level, computing $\mathcal{O}_K^\times$ means finding a $\mathbb{Z}$ -basis of the lattice

$$\iota(\mathcal{O}_K^\times) \subseteq H.$$

In general this is difficult, but in the low-rank cases one can often do it by hand. The main tricks are:

- convert “being a unit” into a norm equation;
- in rank 1, the smallest positive non-torsion unit is automatically fundamental;
- in real quadratic fields, continued fractions solve the relevant norm equation.

The first point is completely general by Theorem 2.18. So the first thing to do in practice is to write down the norm explicitly.

Lemma 5.30 (Rank-1 trick). *Suppose that $\text{rk } \mathcal{O}_K^\times = 1$. Let $\lambda : \mathcal{O}_K^\times \rightarrow \mathbb{R}$ be a homomorphism obtained by composing ι with a coordinate projection, and assume that λ is not identically zero on $\mathcal{O}_K^\times / \mu_K$. If $\epsilon \in \mathcal{O}_K^\times$ satisfies $\lambda(\epsilon) > 0$ and is minimal among all units with positive λ -value, then ϵ is a fundamental unit.*

Proof. By Theorem 4.3,

$$\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}.$$

Since λ is nonzero on the free part, $\lambda(\mathcal{O}_K^\times)$ is a nonzero discrete subgroup of $\mathbb{R}$, hence of the form $\delta\mathbb{Z}$ for some $\delta > 0$. The least positive value in $\lambda(\mathcal{O}_K^\times)$ is therefore δ , and any unit attaining it generates the free part of $\mathcal{O}_K^\times$. $\square$

This already settles the easiest case.

Proposition 5.31 (Imaginary quadratic fields). *Let $K = \mathbb{Q}(\sqrt{d})$ with $d < 0$ squarefree. Then $\mathcal{O}_K^\times = \mu_K$ is finite. More precisely:*

$$\mathcal{O}_K^\times = \begin{cases} \{\pm 1, \pm i\}, & K = \mathbb{Q}(i), \\ \{\pm 1, \pm \omega, \pm \omega^2\}, & K = \mathbb{Q}(\sqrt{-3}), \\ \{\pm 1\}, & \text{otherwise,} \end{cases}$$

where $\omega = \frac{-1+\sqrt{-3}}{2}$.

Proof. By Theorem 4.3, imaginary quadratic fields have unit rank

$$r_1 + r_2 - 1 = 0 + 1 - 1 = 0,$$

so every unit is a root of unity.

If $\zeta \in K$ has order m , then $\mathbb{Q}(\zeta) \subseteq K$, hence

$$\varphi(m) = [\mathbb{Q}(\zeta) : \mathbb{Q}] \leq [K : \mathbb{Q}] = 2.$$

Thus $m \in \{1, 2, 3, 4, 6\}$. The cases $m = 1, 2$ give only ± 1 . If $m = 4$ then $\mathbb{Q}(\zeta) = \mathbb{Q}(i)$. If $m = 3$ or 6 then $\mathbb{Q}(\zeta) = \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$. This proves the classification. $\square$

So imaginary quadratic fields are done. The first interesting case is the real quadratic one.

Proposition 5.32. *Let $K = \mathbb{Q}(\sqrt{d})$ with $d > 1$ squarefree.*

(a) *If $d \equiv 2, 3 \pmod{4}$, then $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$, and for*

$$u = x + y\sqrt{d} \in \mathcal{O}_K$$

one has

$$u \in \mathcal{O}_K^\times \iff x^2 - dy^2 = \pm 1.$$

(b) *If $d \equiv 1 \pmod{4}$, then $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$, and every element of $\mathcal{O}_K$ is of the form*

$$u = \frac{m + n\sqrt{d}}{2} \quad (m \equiv n \pmod{2}).$$

For such a u one has

$$u \in \mathcal{O}_K^\times \iff m^2 - dn^2 = \pm 4.$$

Proof.

(a) The conjugate of $u = x + y\sqrt{d}$ is $\bar{u} = x - y\sqrt{d}$, so

$$N_{K/\mathbb{Q}}(u) = u\bar{u} = x^2 - dy^2.$$

Now apply Theorem 2.18.

(b) The norm is

$$N_{K/\mathbb{Q}}(u) = \frac{m + n\sqrt{d}}{2} \cdot \frac{m - n\sqrt{d}}{2} = \frac{m^2 - dn^2}{4}.$$

Again Theorem 2.18 gives the result. □

Lemma 5.33. *Let K be a real quadratic field, and let $u > 1$ be a unit in the ring of integers $\mathcal{O}_K$. Then $u \geq \phi$, where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.*

Proof. Note u is a root of a monic quadratic polynomial with integer coefficients:

$$x^2 - Tx + N = 0$$

where $T = \text{Tr}_{K/\mathbb{Q}}(u) = u + \bar{u}$ and $N = u\bar{u} = \pm 1$. Since $u > 1$ and $N = \pm 1$, its conjugate $\bar{u} = \frac{N}{u}$ strictly satisfies $-1 < \bar{u} < 1$. Now

$$T = u + \bar{u} > 1 - 1 = 0$$

but since T must be a strictly positive integer, we know $T \geq 1$.

Using the quadratic formula, and knowing $u > 1$, we take the positive root:

$$u = \frac{T + \sqrt{T^2 - 4N}}{2}$$

To find the absolute minimum value for u , we evaluate the two possible cases for the norm N :

Case 1, $N = 1$: The equation becomes $u = \frac{T + \sqrt{T^2 - 4}}{2}$. For u to be real and strictly greater than 1, we must have $T^2 - 4 > 0$. The smallest integer trace satisfying this condition is $T = 3$. This yields:

$$u = \frac{3 + \sqrt{3^2 - 4}}{2} = \frac{3 + \sqrt{5}}{2}$$

Because $\frac{3+\sqrt{5}}{2} > \phi$, this case does not produce the minimum. Comparing both cases, the absolute smallest possible unit $u > 1$ across all real quadratic fields is exactly ϕ .

Case 2, $N = -1$: The equation becomes $u = \frac{T + \sqrt{T^2 + 4}}{2}$. Since the term inside the square root is always positive, we minimize u by choosing the absolutely smallest allowable integer trace, $T = 1$. This yields:

$$u = \frac{1 + \sqrt{1^2 + 4}}{2} = \frac{1 + \sqrt{5}}{2} = \phi$$

□

Corollary 5.34. *In a real quadratic field, the smallest positive solution of the relevant norm equation in Proposition 5.32 gives a fundamental unit.*

Proof. Suppose that u is a fundamental unit. Since

$$N(u) = u \cdot \bar{u} = \pm 1 \implies |u| \cdot |\bar{u}| = 1$$

we may assume that $u > 1$ (otherwise swap with $\bar{u}$). Moreover its conjugate is $\bar{u} = \pm u^{-1}$. Because $u > 1$, it strictly follows that $\bar{u} \in (-1, 1)$.

By Dirichlet's Unit Theorem, any unit $v > 1$ in this field is a positive power of the fundamental unit, meaning $v = u^n$ for some integer $n \geq 1$. We can isolate the coefficients x_n and y_n for any such unit u^n :

$$x_n = u^n + \bar{u}^n \quad \text{and} \quad y_n \sqrt{d} = u^n - \bar{u}^n$$

Since $u > 1$, its powers u^n are strictly greater than 1. Likewise, since $\bar{u} \in (-1, 1)$, its powers $\bar{u}^n \in (-1, 1)$. This guarantees that for all $n \geq 1$, both $x_n > 0$ and $y_n > 0$.

Substituting $\bar{u}^n = (\pm 1)^n u^{-n}$, we can express the first coefficient x_n explicitly as a function of the exponent n :

$$x_n = u^n + (\pm 1)^n u^{-n} \quad y_n \sqrt{d} = u^n - (\pm 1)^n u^{-n}$$

Therefore

$$x_{n+1} - x_n = \begin{cases} (u-1)(u^n - u^{-n-1}) & \text{if } \bar{u} = u^{-1} \\ u^n(u-1) - (-1)^n u^{-n-1}(u+1) & \text{if } \bar{u} = -u^{-1} \end{cases}$$

$$\sqrt{d}(y_{n+1} - y_n) = \begin{cases} (u-1)(u^n + u^{-n-1}) & \text{if } \bar{u} = u^{-1} \\ u^n(u-1) + (-1)^n u^{-n-1}(u+1) & \text{if } \bar{u} = -u^{-1} \end{cases}$$

It is clear that in the first case x_n and y_n are both monotonically increasing since $u > 1 > u^{-1}$. In the other case we want

$$u^{2n+1}(u-1) > (u+1)$$

in order for x_n and y_n to monotonically increase for all n . Since the RHS is increasing it suffices to check $n = 1$:

$$u^3(u-1) > (u+1)$$

We use Lemma 5.33 (u is greater than the golden ratio). In the case that the equality is achieved (which can only happen when $K = \mathbb{Q}[\sqrt{5}]$) we can only show that x_n is strictly increasing and y_n is non-decreasing. Otherwise, our inequality is attained and both x_n and y_n must strictly increase. $\square$

This provides a general method for real quadratic fields: test positive x and y as solutions to the norm equation until a solution is found. Once a solution is found, we need only exhaustively check strictly smaller positive values of x, y to guarantee we have found the fundamental unit, which is a finite and manageable task.

Remark 5.35 (A practical trick in any rank-1 field). The real quadratic case is the cleanest, but Lemma 5.30 is more general. If K has unit rank 1 (for instance a complex cubic field), then after writing down an integral basis of $\mathcal{O}_K$ one can simply test small algebraic integers α by the norm criterion

$$N_{K/\mathbb{Q}}(\alpha) = \pm 1.$$

Once one finds a non-torsion unit, the smallest one under any nonzero logarithmic coordinate is automatically fundamental. This is often the fastest by-hand computation in simple examples. There is a faster algorithm by computing *Pell-convergents*. I will define the algorithm and state why it works, but I will not prove a lot of things.

Definition 5.36 (Continued Fractions). We define recursively

$$[a_0] = a_0 \quad [a_0; a_1, a_2, \dots, a_n] = a_0 + \frac{1}{[a_1; a_2, \dots, a_n]}$$

which gives a good notation to express a continued fraction.

Definition 5.37 (Pell convergents). Write

$$\sqrt{d} = [a_0; a_1, a_2, \dots], \quad \frac{p_n}{q_n} = [a_0; a_1, \dots, a_n].$$

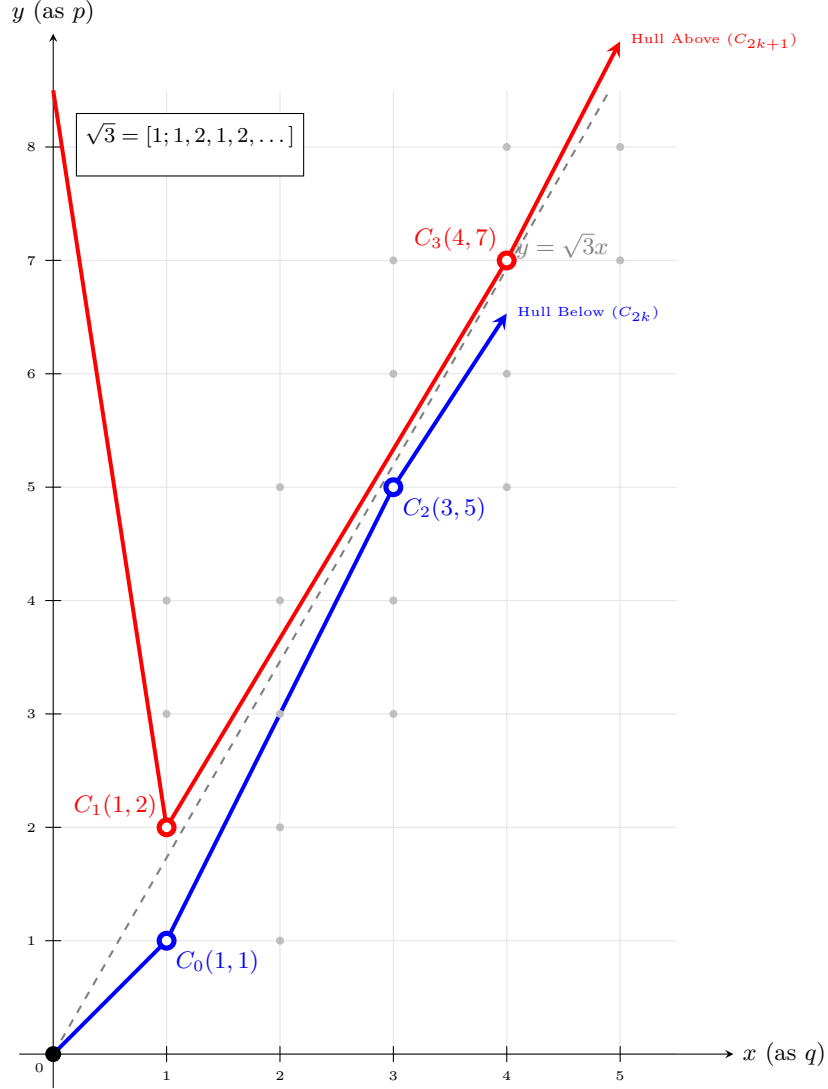
The fractions p_n/q_n are the *Pell convergents* of $\sqrt{d}$.

Theorem 5.38. *The Pell-convergents are the successive best rational approximations to $\sqrt{d}$. Explicitly*

$$\left| \sqrt{d} - \frac{p_n}{q_n} \right| < \left| \sqrt{d} - \frac{a}{b} \right| \quad \text{for every } 0 < b \leq q_n \text{ with } \frac{a}{b} \neq \frac{p_n}{q_n}.$$

and every fraction which satisfies this is a Pell-convergent.

Geometrically, one plots the lattice points (q, p) and the line $y = \sqrt{d}x$. So in the Pell equation $p^2 - dq^2 = \pm 1$, the ratio p/q should be thought of as the lattice points trying to hug the irrational slope $\sqrt{d}$ as tightly as possible.



Proposition 5.39 (Why Pell solutions are convergents). *Assume $d \equiv 2, 3 \pmod{4}$. If $p, q > 0$ satisfy*

$$p^2 - dq^2 = \pm 1,$$

then p/q is a Pell convergent of $\sqrt{d}$.

Proof. Note that:

$$\frac{p^2}{q^2} = d \pm \frac{1}{q^2} \implies \frac{p}{q} = \sqrt{d \pm \frac{1}{q^2}}$$

More generally

$$\frac{a}{b} = \sqrt{d + \frac{a^2 - db^2}{b^2}}$$

and since the numerator $a^2 - db^2$ is at least one (or four in the other case), a lot of bounding shows that p/q is the best rational approximation with denominator less than q . Therefore it is a Pell-convergent. $\square$

This is why continued fractions solve the unit problem in the real quadratic case: once the norm equation has been reduced to Pell, one no longer searches blindly through all lattice points. One walks through the pell-convergents and checks whether the norm equation is satisfied. Since the Pell-convergents increase in size exponentially, we check far fewer points and this is a significant speedup.

Remark 5.40. To compute the Pell convergents for $\sqrt{d}$, we first determine the continued fraction expansion $\sqrt{d} = [a_0; \overline{a_1, a_2, \dots, a_k}]$.

The coefficients a_n are generated using the recursive auxiliary sequences P_n and Q_n :

$$\begin{aligned} P_0 &= 0, & Q_0 &= 1, & a_0 &= \lfloor \sqrt{d} \rfloor \\ P_n &= a_{n-1}Q_{n-1} - P_{n-1} \\ Q_n &= \frac{d - P_n^2}{Q_{n-1}} \\ a_n &= \left\lfloor \frac{a_0 + P_n}{Q_n} \right\rfloor \end{aligned}$$

The numerators p_n and denominators q_n of the convergents are defined recursively by:

$$\begin{aligned} p_{-2} &= 0, & p_{-1} &= 1, & p_n &= a_n p_{n-1} + p_{n-2} \\ q_{-2} &= 1, & q_{-1} &= 0, & q_n &= a_n q_{n-1} + q_{n-2} \end{aligned}$$

Equivalently, this recurrence can be expressed and computed efficiently via matrix multiplication. The convergents are obtained by the product of 2×2 matrices corresponding to the continued fraction coefficients:

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix}$$

By repeatedly computing the matrix product, the first column of the resulting matrix directly yields the numerator and denominator of the convergent $\frac{p_n}{q_n}$.

6. Cyclotomic Fields

Cyclotomic fields are extremely useful, in this section we will understand them.

6.1. The Ring of Integers

If I asked you to guess what the ring of integers of $\mathbb{Q}[\zeta_n]$ is you would probably say $\mathbb{Z}[\zeta_n]$. This is completely correct, but proving it requires some work. Nevertheless, using the tools of Section 5.3 it is quite straightforward, so one might try to do it as an exercise.

Problem 6.1. If p is an odd prime then the minimal polynomials of ζ_p and ζ_{p^k} are respectively:

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + 1 \quad \Phi_{p^k}(x) = \Phi_p(x^{p^{k-1}})$$

Lemma 6.2. Let p be an odd prime and Φ_p be the p -th cyclotomic polynomial. The discriminant of Φ_p is given by

$$\Delta_K = (-1)^{\frac{p(p-1)}{2}} p^{p-2}$$

Proof. We use the formula:

$$\Delta_K = (-1)^{\frac{n(n-1)}{2}} N_{K/\mathbb{Q}}(\Phi'_p(\zeta_p))$$

Note that

$$\begin{aligned}
(x-1)\Phi_p(x) &= x^p - 1 && \text{Problem 6.1} \\
\Phi_p(x) + (x-1)\Phi'_p(x) &= px^{p-1} && \text{differentiate both sides} \\
(\zeta_p - 1)\Phi'_p(\zeta_p) &= p\zeta_p^{p-1} && \text{evaluate at } \zeta_p \\
N(\zeta_p - 1)N(\Phi'_p(\zeta_p)) &= N(p)N(\zeta_p^{p-1}) && \text{take norm on both sides}
\end{aligned}$$

We evaluate the component norms as follows:

- $N(p) = p^{p-1}$, since $p \in \mathbb{Q}$ and the extension degree is $p-1$.
- $N(\zeta_p^{p-1}) = 1$, because ζ_p is a root of unity.
- $\Phi_p(1) = \prod_{i=1}^{p-1} (1 - \zeta_p^i) = p$, which implies $N(1 - \zeta_p) = p$. Because $p-1$ is even for odd primes, $N(\zeta_p - 1) = N(1 - \zeta_p) = p$.

Substituting these values into our norm equation, we find:

$$p \cdot N(\Phi'_p(\zeta_p)) = p^{p-1} \implies N(\Phi'_p(\zeta_p)) = p^{p-2}$$

Finally, we substitute this back into the discriminant formula. The sign exponent is $\frac{(p-1)(p-2)}{2}$. For any odd prime p , $\frac{(p-1)(p-2)}{2}$ shares the same parity as $\frac{p(p-1)}{2}$. Therefore:

$$\Delta_K = (-1)^{\frac{p(p-1)}{2}} p^{p-2}$$

□

Lemma 6.3. *Let $K = \mathbb{Q}(\alpha)$ be a number field of degree n , where $\alpha \in \mathcal{O}_K$ is a root of a monic polynomial $f(x) \in \mathbb{Z}[x]$ that is Eisenstein at a prime p . Then p does not divide the index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$.*

Proof. We might apply Dedekind's criterion on $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$. Note that

$$f(x) \equiv x^n \pmod{p} \quad f(x) = x^n + ph(x) \quad h(x) := \frac{a_{n-1}}{p} x^{n-1} + \dots + \frac{a_0}{p}$$

where $h(x) \in \mathbb{Z}[x]$ and $a_0/p \not\equiv 0 \pmod{p}$ by the Eisenstein assumption. Therefore $\bar{x}$ does not divide $\bar{h}(x)$ and we are done by Theorem 5.24. □

Theorem 6.4. *Let p be prime and $K = \mathbb{Q}(\zeta_p)$, where ζ_p is a primitive p -th root of unity. The ring of integers of K is $\mathcal{O}_K = \mathbb{Z}[\zeta_p]$.*

Proof. Let $\lambda = \zeta_p - 1$. Since $\zeta_p \in \mathbb{Z}[\lambda]$ and $\lambda \in \mathbb{Z}[\zeta_p]$, we have $\mathbb{Z}[\zeta_p] = \mathbb{Z}[\lambda]$. We will show that $\mathcal{O}_K = \mathbb{Z}[\lambda]$. The minimal polynomial of ζ_p over $\mathbb{Q}$ is the cyclotomic polynomial:

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + 1$$

The minimal polynomial of λ is therefore $f(x) = \Phi_p(x+1)$, which evaluates to:

$$f(x) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{1} x^{p-2} + \dots + \binom{p}{p-1}$$

Notice that $f(x)$ is a monic polynomial of degree $p-1$. For $1 \leq k \leq p-1$, the binomial coefficient $\binom{p}{k}$ is divisible by p . The constant term is $\binom{p}{p-1} = p$, which is divisible by p but not by p^2 . Thus, $f(x)$ is Eisenstein at p . By the previous Lemma, since λ is a root of a polynomial that is Eisenstein at p , we know that:

$$p \nmid [\mathcal{O}_K : \mathbb{Z}[\lambda]]$$

however p is the only prime dividing the discriminant of $\mathbb{Z}[\lambda]/\mathbb{Z}$ by Lemma 6.2, so we are done. □

Problem 6.5. Convince yourself that the methods used generalize to $\mathcal{O}_{\mathbb{Q}(\zeta_{p^k})} = \mathbb{Z}[\zeta_{p^k}]$ where p is an odd prime.

Theorem 6.6. For any $n \geq 1$, the ring of integers of $K = \mathbb{Q}(\zeta_n)$ is $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$.

Proof. We proceed by induction on the number of distinct prime factors of n .

Base Case: If $n = p^k$, then $\mathcal{O}_K = \mathbb{Z}[\zeta_{p^k}]$ by Problem 6.5.

Inductive Step: Let $n = m \cdot p^k$ with $\gcd(m, p) = 1$. Let $K = \mathbb{Q}(\zeta_m)$ and $L = \mathbb{Q}(\zeta_{p^k})$. By the inductive hypothesis, $\mathcal{O}_K = \mathbb{Z}[\zeta_m]$ and $\mathcal{O}_L = \mathbb{Z}[\zeta_{p^k}]$. Since the primes dividing Δ_K must divide m , and the primes dividing d_L must divide p (by the tower law for discriminants), we have $\gcd(\Delta_K, \Delta_L) = 1$. Now by Theorem 5.29:

$$\mathcal{O}_{\mathbb{Q}(\zeta_n)} = \mathcal{O}_{KL} = \mathcal{O}_K \mathcal{O}_L = \mathbb{Z}[\zeta_m] \mathbb{Z}[\zeta_{p^k}] = \mathbb{Z}[\zeta_n]$$

□

7. The Galois Case

7.1. Decomposition and Inertia Groups

Assume from now on that L/K is a finite *Galois* extension and write

$$G := \text{Gal}(L/K).$$

Let $\mathfrak{p} \subset R$ be a nonzero prime and choose a prime $\mathfrak{q} \subset S$ with $\mathfrak{q} \cap R = \mathfrak{p}$. Write $\kappa(\mathfrak{p}) := R/\mathfrak{p}$ and $\kappa(\mathfrak{q}) := S/\mathfrak{q}$ for the residue fields.

Lemma 7.1. G acts transitively on the primes above $\mathfrak{p}$.

Proof. Suppose FTSOC that there are two distinct G orbits of primes above $\mathfrak{p}$. Call one of them them $\{\mathfrak{q}_1, \dots, \mathfrak{q}_n\}$ and let $\{\mathfrak{r}_i\}$ be the primes above $\mathfrak{p}$ not in this orbit. Using CRT we may find $x \in S$ with:

$$x \equiv 1 \pmod{\mathfrak{q}_i} \quad x \equiv 0 \pmod{\mathfrak{r}_i}$$

Note that $y = \prod_{\sigma \in G} \sigma(x)$ also has these residues and is an element of R . Therefore

$$y \in (R \cap \mathfrak{q}_1) = \mathfrak{p}$$

which is a contradiction with the fact that $y \notin \mathfrak{q}_1$, since it is an ideal that contains $\mathfrak{p}$. □

Corollary 7.2. Every prime above $\mathfrak{p}$ has the same ramification index and inertia degree. Therefore Theorem 2.15 simplifies to

$$n = efg$$

where g is the number of primes above $\mathfrak{p}$.

Definition 7.3 (Decomposition group).

- The *decomposition group* of $\mathfrak{q}$ (over $\mathfrak{p}$) is the stabilizer

$$D_{\mathfrak{q}} := \{\sigma \in G : \sigma(\mathfrak{q}) = \mathfrak{q}\} \subseteq G.$$

- Every $\sigma \in D_{\mathfrak{q}}$ induces a $\kappa(\mathfrak{p})$ -automorphism of $\kappa(\mathfrak{q})$, hence a homomorphism:

$$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \longrightarrow \text{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

The *inertia group* is the kernel:

$$I_{\mathfrak{q}} := \ker(\rho_{\mathfrak{q}}) = \{\sigma \in D_{\mathfrak{q}} : \sigma(x) \equiv x \pmod{\mathfrak{q}} \text{ for all } x \in S\}$$

Proposition 7.4. *The reduction map $\rho_{\mathfrak{q}}$ fits into a short exact sequence*

$$1 \longrightarrow I_{\mathfrak{q}} \longrightarrow D_{\mathfrak{q}} \xrightarrow{\rho_{\mathfrak{q}}} \text{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p})) \longrightarrow 1$$

Moreover the extension $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ is Galois and has Galois group $\text{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$ by Proposition 7.4.

Proof. Firstly, $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ is Galois since we defined $I_{\mathfrak{q}}$ without making any choices. The main task is to show that $\rho_{\mathfrak{q}}$ is surjective. Because $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is finite separable we may choose a primitive element $\bar{\alpha}$ with $\kappa(\mathfrak{q}) = \kappa(\mathfrak{p})[\bar{\alpha}]$. By Chinese remainder theorem we may find an element α with

$$\alpha \equiv \bar{\alpha} \pmod{\mathfrak{q}} \quad \alpha \equiv 0 \pmod{\mathfrak{r}_i} \quad (1)$$

where $\mathfrak{r}_i$ are all the other primes lying over $\mathfrak{p}$. Let

$$f(x) = \prod_{\sigma \in G} (x - \sigma(\alpha)) \quad \bar{f}(x) = \prod_{\sigma \in G} (x - \overline{\sigma(\alpha)})$$

be the characteristic polynomial of α and its reduction in the residue extension respectively. Since f has coefficients in R , $\bar{f}$ has coefficients in $R/\mathfrak{p}$. Therefore any $\tau \in \text{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$ sends $\bar{\alpha}$ to another root of $\bar{f}(x)$ and:

$$\tau(\bar{\alpha}) = \overline{\sigma(\alpha)} \quad \text{for some } \sigma \in G$$

We claim that $\rho_{\mathfrak{q}}(\sigma) = \tau$.

- Suppose for the sake of contradiction that $\sigma \notin D_{\mathfrak{q}}$. Then by Eq. (1) $\sigma(\alpha) \equiv 0 \pmod{\mathfrak{q}}$. Therefore $\overline{\sigma(\alpha)}$ is 0 in the residue extension, a contradiction since τ can't send a nonzero element to 0.
- Since α is primitive, $\tau(\bar{\alpha}) = \overline{\sigma(\alpha)}$ forces $\rho_{\mathfrak{q}}(\sigma_k) = \tau$.

□

Remark 7.5 (The efg -formula realized by subextensions).

- (a) $|D_{\mathfrak{q}}| = ef$, $|I_{\mathfrak{q}}| = e$, and $|D_{\mathfrak{q}}/I_{\mathfrak{q}}| = f$.
- (b) Considering fixed fields

$$K \subseteq L^{D_{\mathfrak{q}}} \subseteq L^{I_{\mathfrak{q}}} \subseteq L$$

and the degrees in this tower recover the splitting invariants:

$$[L^{D_{\mathfrak{q}}} : K] = g, \quad [L^{I_{\mathfrak{q}}} : L^{D_{\mathfrak{q}}}] = f, \quad [L : L^{I_{\mathfrak{q}}}] = e.$$

Proof.

- (a) By orbit stabilizer theorem $|D_{\mathfrak{q}}| = |G|/g$. Then plugging in $|G| = efg$ gives $|D_{\mathfrak{q}}| = ef$ as desired. We must have:

$$ef = |D_{\mathfrak{q}}| = |I_{\mathfrak{q}}| |D_{\mathfrak{q}}/I_{\mathfrak{q}}|$$

Note by Proposition 7.4 that

$$|D_{\mathfrak{q}}/I_{\mathfrak{q}}| = [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})] = f$$

which forces $|I_{\mathfrak{q}}| = e$.

- (b) This is now a direct translation via the Galois correspondence:

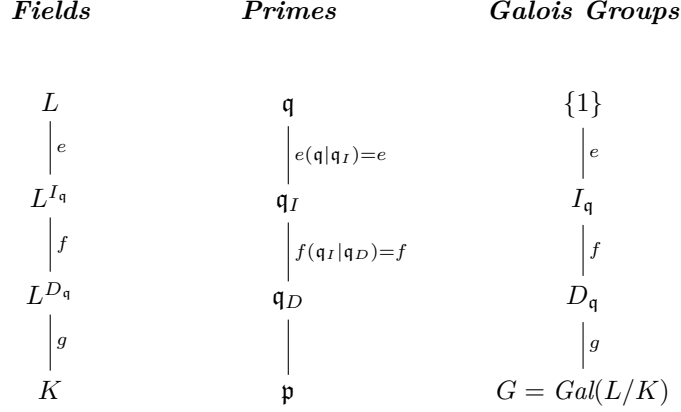
$$[L^{D_{\mathfrak{q}}} : K] = [G : D_{\mathfrak{q}}] = g, \quad [L^{I_{\mathfrak{q}}} : L^{D_{\mathfrak{q}}}] = [D_{\mathfrak{q}} : I_{\mathfrak{q}}] = f, \quad [L : L^{I_{\mathfrak{q}}}] = |I_{\mathfrak{q}}| = e.$$

□

Theorem 7.6 (How to read the tower). *Define:*

$$\mathfrak{q}_I = \mathcal{O}_{L^{I_{\mathfrak{q}}}} \cap \mathfrak{q} \quad \mathfrak{q}_D = \mathcal{O}_{L^{D_{\mathfrak{q}}}} \cap \mathfrak{q}$$

Then we can draw the diagram



where the inertia and ramification degrees are assumed 1 unless indicated.

Proof. We will make some counting arguments with Theorem 2.15.

- Consider the extension $L/L^{D_{\mathfrak{q}}}$. Since the Galois group of this extension fixes $\mathfrak{q}$ there are no other primes above $\mathfrak{q}_D$ by Lemma 7.1. Therefore by Corollary 7.2 $\mathfrak{q}_D$ does not split in this extension. Now since $L/L^{D_{\mathfrak{q}}}$ has degree ef we must have $f(\mathfrak{q}|\mathfrak{q}_D) = f$ and $e(\mathfrak{q}|\mathfrak{q}_D) = e$.
- Consider the extension $L/L^{I_{\mathfrak{q}}}$. By definition the Galois group acts trivially on the residue field extension. On the other hand the Galois group surjects onto the residue field automorphism group by the argument in Proposition 7.4. Therefore this extension must have no inertia since the residue field has no nontrivial automorphism. Then $e(\mathfrak{q}|\mathfrak{q}_D) = e$ by (a).
- By the previous results $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ must account for the inertia. Moreover by counting degrees $f(\mathfrak{q}_I|\mathfrak{q}_D) = f$.

□

Corollary 7.7. *In the case that $L^{D_{\mathfrak{q}}}/K$ is Galois (or equivalently $D_{\mathfrak{q}}$ is normal)*

- $\mathfrak{p}$ is totally split in $L^{D_{\mathfrak{q}}}/K$;
- every prime over $\mathfrak{p}$ is inert in $L^{D_{\mathfrak{q}}}/L^{I_{\mathfrak{q}}}$;
- every prime over $\mathfrak{p}$ is totally ramified in $L^{D_{\mathfrak{q}}}/L^{I_{\mathfrak{q}}}$.

Theorem 7.8. *Suppose that $K \subset E \subset L$ is an intermediate extension with $\text{Gal}(L/E) := H \subset G$. Then*

$$I_{\mathfrak{q}}(L/E) = I_{\mathfrak{q}}(L/K) \cap H \quad D_{\mathfrak{q}}(L/E) = D_{\mathfrak{q}}(L/K) \cap H$$

and moreover if $\mathfrak{q}_E = \mathfrak{q} \cap E$ then there is the following commutative diagram of exact sequences:

$$\begin{array}{ccccccc}
& & 1 & & 1 & & 1 \\
& & \downarrow & & \downarrow & & \downarrow \\
1 & \longrightarrow & I_{\mathfrak{q}}(L/E) & \longrightarrow & I_{\mathfrak{q}}(L/K) & \longrightarrow & I_{\mathfrak{q}_E}(E/K) \longrightarrow 1 \\
& & \downarrow & & \downarrow & & \downarrow \\
1 & \longrightarrow & D_{\mathfrak{q}}(L/E) & \longrightarrow & D_{\mathfrak{q}}(L/K) & \longrightarrow & D_{\mathfrak{q}_E}(E/K) \longrightarrow 1 \\
& & \downarrow & & \downarrow & & \downarrow \\
1 & \longrightarrow & \overline{G}_{\mathfrak{q}}(L/E) & \longrightarrow & \overline{G}_{\mathfrak{q}}(L/K) & \longrightarrow & \overline{G}_{\mathfrak{q}_E}(E/K) \longrightarrow 1 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & 1 & & 1 & & 1
\end{array}$$

where we have used $\overline{G}$ to denote the automorphisms of the respective residue field.

Proof. Recall the inertial and decomposition groups are defined as subsets of the Galois group without reference to the base field. Therefore restricting to a subset of the Galois group just intersects.

Now consider the commutative diagram. Each column follows from Proposition 7.4. Each row corresponds to an inclusion and then a restriction, in which the inclusion is precisely the kernel of the restriction by definition. The surjectivity roughly comes from the fact that we can embed a field into a larger field in stages. This gives the following bijection

$$\phi : \text{Hom}_K(L, \Omega) \longleftrightarrow \text{Hom}_E(L, \Omega) \times \text{Hom}_K(E, \Omega)$$

which we can view as a bijection of Galois groups, so that the restriction becomes projection onto the second term on the RHS. If $\sigma \in \text{Gal}(E/K)$ stabilizes $\mathfrak{q}_E$ then $\phi^{-1}(1, \sigma) \in \text{Gal}(L, K)$ stabilizes $\mathfrak{q}$ and restricts to σ . This shows surjectivity on the first two rows, and the third follows either from a similar argument, or the nine lemma. $\square$

Remark 7.9. The most natural way to remember this is that:

- $L^{D_{\mathfrak{q}}}$ is a largest subextension of L/K such that $e(\mathfrak{q} \cap L^{D_{\mathfrak{q}}} | \mathfrak{p}) = f(\mathfrak{q} \cap L^{D_{\mathfrak{q}}} | \mathfrak{p}) = 1$.
- $L^{I_{\mathfrak{q}}}$ is the largest subextension of L/K such that $e(\mathfrak{q} \cap L^{I_{\mathfrak{q}}} | \mathfrak{p}) = 1$.

Proof. Suppose that $K \subset K' \subset L$ is an intermediate extension and consider L/K' . Theorem 7.8 tells us that the inertial and decomposition groups are intersections with $\text{Gal}(L/K')$. Since group intersection corresponds to field multiplication in the Galois correspondence we obtain the diagram

$$\begin{array}{ccc}
L & & \\
e \downarrow & \searrow e' & \\
L^{I_{\mathfrak{q}}} & & L^{I_{\mathfrak{q}}} K' \\
f \downarrow & \nearrow f' & \downarrow f' \\
L^{D_{\mathfrak{q}}} & & L^{D_{\mathfrak{q}}} K' \\
g \downarrow & \nearrow g' & \downarrow g' \\
K & & K'
\end{array}$$

Since e, f, g are multiplicative:

- K'/K being totally split is exactly that $e' = e$ and $f' = f$. This occurs when $L^{D_{\mathfrak{q}}}K' = L^{D_{\mathfrak{q}}}$ or equivalently $K' \in L^{D_{\mathfrak{q}}}$.
- K'/K being unramified is exactly that $e' = e$. This occurs when $L^{I_{\mathfrak{q}}}K' = L^{I_{\mathfrak{q}}}$ or equivalently $K' \in L^{I_{\mathfrak{q}}}$.

□

While this stuff is extremely beautiful, it is also very useful even when things aren't Galois. For instance consider how this understanding enables the following argument.

Theorem 7.10. *Let E_1 and E_2 be extensions of K . Then:*

- *If $\mathfrak{p}$ is unramified in E_1/K and E_2/K it is unramified in the compositum E_1E_2/K .*
- *If $\mathfrak{p}$ is totally split in E_1/K and E_2/K it is totally split in the compositum E_1E_2/K .*

Proof. Let L be the normal closure of E_1E_2 so that L/K is Galois. By Remark 7.9:

- If $\mathfrak{p}$ is unramified in both extensions then $E_1, E_2 \subseteq L^{I_{\mathfrak{q}}}$ for every prime $\mathfrak{q}$ above $\mathfrak{p}$ in L . Therefore $E_1E_2 \subseteq L^{I_{\mathfrak{q}}}$ exactly meaning $\mathfrak{p}$ is unramified in the compositum.
- If $\mathfrak{p}$ is totally split in both extensions then $E_1, E_2 \subseteq L^{D_{\mathfrak{q}}}$ for every prime $\mathfrak{q}$ above $\mathfrak{p}$ in L . Therefore $E_1E_2 \subseteq L^{D_{\mathfrak{q}}}$ exactly meaning $\mathfrak{p}$ is totally split in the compositum.

□

Corollary 7.11. *A prime $\mathfrak{p}$ is ramified (resp not totally split) in an extension E/K if and only if it is ramified (resp totally split) in the Galois closure of E .*

Proof. The Galois closure of a field is the compositum of its unique embeddings. Now apply Theorem 7.10. □

Remark 7.12. Suppose that $K = \mathbb{Q}$. Then the residue field $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is of the form $\mathbb{F}_{p^n}/\mathbb{F}_p$ for some prime p . Now

$$D_{\mathfrak{q}}/I_{\mathfrak{q}} \cong \text{Gal}(L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}) \cong \text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$$

is a cyclic group of order n generated by the *Frobenius element* $x \mapsto x^p$ on the right hand side. In particular when $\mathfrak{p}$ is unramified then $I_{\mathfrak{q}} = 1$ and we may find a unique element $\sigma \in D_{\mathfrak{q}}$ with

$$\sigma(\alpha) \equiv \alpha^p \pmod{\mathfrak{q}}$$

The Frobenius element is extremely useful, and will make many appearances when studying more number theory.

Problem 7.13. Suppose that $K \subseteq E \subseteq L$ is an intermediate field with $\text{Gal}(L/E) = H$. Then there is a bijective correspondence:

$$\text{primes above } \mathfrak{p} \longleftrightarrow \text{double cosets } H\sigma D_{\mathfrak{q}}$$

The Frobenius element is defined only with respect to a single prime over $\mathfrak{p}$. However the next theorem claims it somehow "knows" something about the other primes over $\mathfrak{p}$. This is very non-intuitive but interesting.

Theorem 7.14. *Suppose that $\mathbb{Q} \subseteq E$ and let L be the normal closure of E . The Frobenius element has a natural action on the embeddings of E into L . If p is unramified in E then the cycle type of the Frobenius element is exactly the inertial degrees of the primes above p in E .*

Proof. Let $H := \text{Gal}(L/E)$ and note that the embeddings of E into L correspond to the left cosets σH . The Frobenius element is characterized by being a generator of D_q , therefore the orbits of the embeddings under the Frobenius element correspond to the double cosets $D_q \sigma H$. We have the natural map

$$D_q \backslash G/H \rightarrow H \backslash G/D_q \quad D_q \sigma H \mapsto H \sigma^{-1} D_q$$

which associates Frobenius element orbits to primes in H above p by Problem 7.13. The size of each orbit is

$$[D_q : D_q \cap \sigma H \sigma^{-1}] = \frac{|D_q|}{|D_q \cap \sigma H \sigma^{-1}|} = \frac{[D_q : I_q]}{[D_q \cap \sigma H \sigma^{-1} : I_q]} = \frac{[\sigma D_q \sigma^{-1} : \sigma I_q \sigma^{-1}]}{[\sigma D_q \sigma^{-1} \cap H : \sigma I_q \sigma^{-1}]}$$

where we have used the fact that the inertial group is contained in any conjugate to H by Corollary 7.11 and Remark 7.9 since p is unramified. The numerator is exactly $f(\sigma(q)/p)$. Since $\sigma D_q \sigma^{-1} \cap H$ is the decomposition group of $\sigma(q)$ over E , the denominator is $f(\sigma(q)|\sigma(q) \cap E)$. By the tower law of inertial degree, the right hand side simplifies to $f(\sigma(q)|p)$ as desired. $\square$

As an extra treat, here is a really nice proof of quadratic reciprocity using this theory.

Theorem 7.15 (Reciprocity). *Let p and q be distinct odd primes. Then:*

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}$$

Proof. Let $K = \mathbb{Q}[\zeta_p]$ where ζ_p is a primitive p th root of unity. Since

$$\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times \cong \mathbb{Z}/(p-1)\mathbb{Z}$$

it contains a unique subgroup of order $(p-1)/2$ containing the elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ that are squares. Therefore K contains a unique quadratic extension F . On the other hand, only p is ramified in $K/\mathbb{Q}$, so F must have this same property. Now Problem 2.17 allows us to deduce F :

- If $p \equiv 1 \pmod{4}$ then p is the only prime ramifying in $\mathbb{Q}[\sqrt{p}]$ and this is the only quadratic field for which this is true.
- If $p \equiv 3 \pmod{4}$ then $-p$ is the only prime ramifying in $\mathbb{Q}[\sqrt{-p}]$ and this is the only quadratic field for which this is true.

Therefore $F = \mathbb{Q}[\sqrt{d}]$ where $d = (-1)^{(p-1)/2} \cdot p$. Since the Galois group of F has order 2 we identify it with $\{+1, -1\}$. Set q to be an odd prime not equal to p and $\tau \in \text{Gal}(K/\mathbb{Q}) = \{+1, -1\}$ to be the Frobenius element with:

$$\tau \alpha \equiv \alpha^q \pmod{q} \implies \tau : \zeta \mapsto \zeta^q$$

Therefore τ restricts to the identity element of $\text{Gal}(F/\mathbb{Q})$ according to whether q is a square in $(\mathbb{Z}/p\mathbb{Z})^\times$ or not. This exactly means $\tau = \left(\frac{q}{p}\right)$. On the other hand the Frobenius element generates the inertial group, which is nontrivial in our case if and only if q remains inert in F . This corresponds to $\tau = \left(\frac{d}{q}\right)$ again by Problem 2.17. Now by equating

$$\left(\frac{q}{p}\right) = \left(\frac{-1}{q}\right)^{(p-1)/2} \cdot \left(\frac{p}{q}\right) = (-1)^{\frac{(p-1)(q-1)}{4}} \cdot \left(\frac{p}{q}\right)$$

where we have used that -1 is a square iff $q \equiv 1 \pmod{4}$. $\square$

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